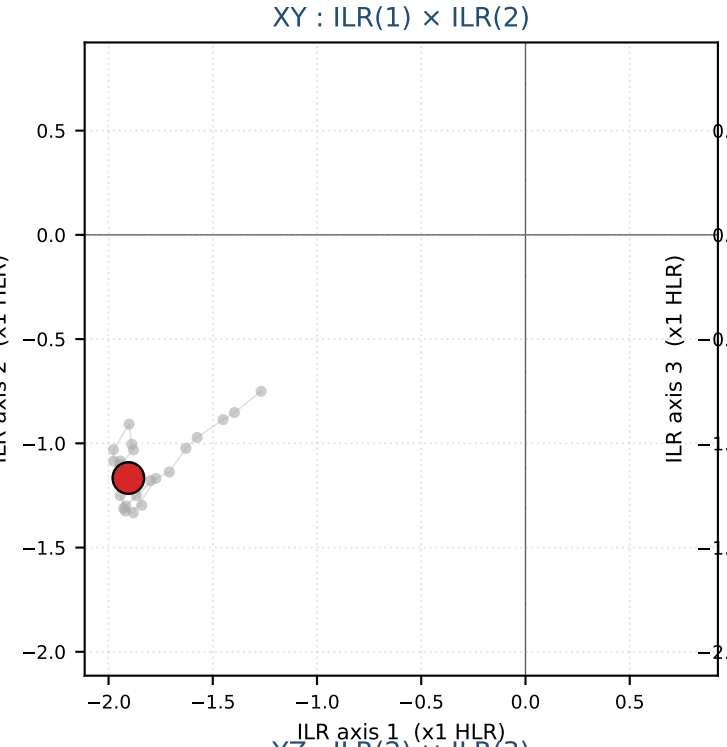
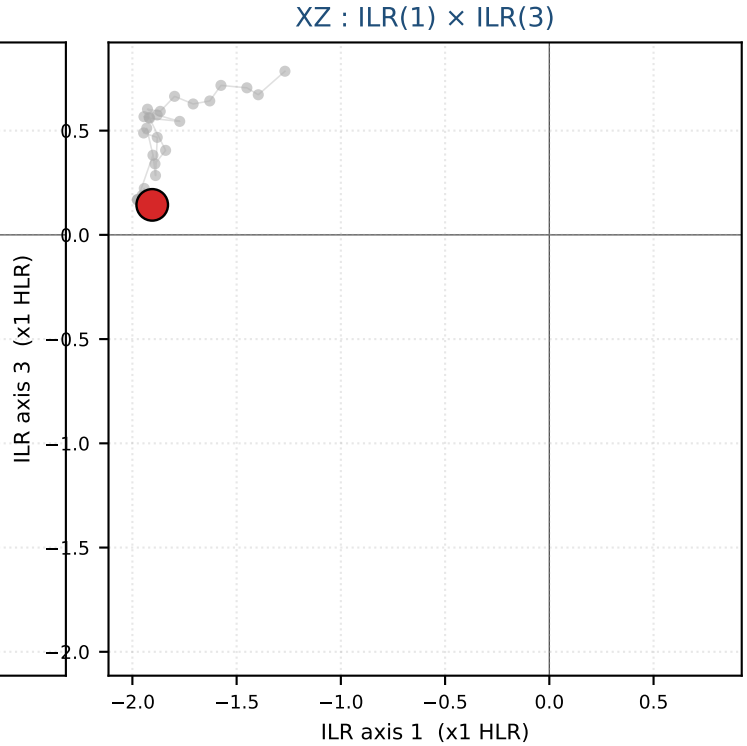
STAGE 1 v4 | t=0 | 2000 | D=8 T=26 | level 1 first-principles

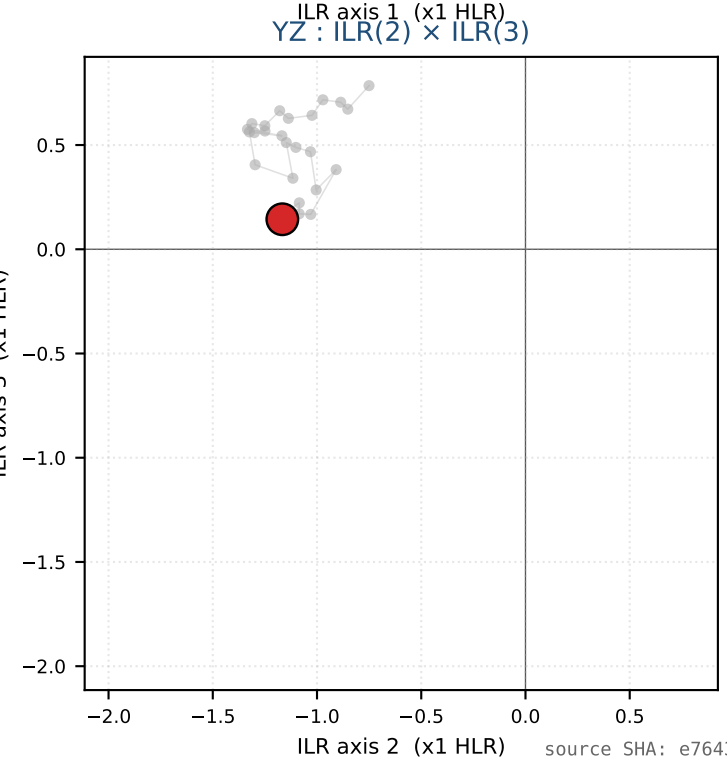
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

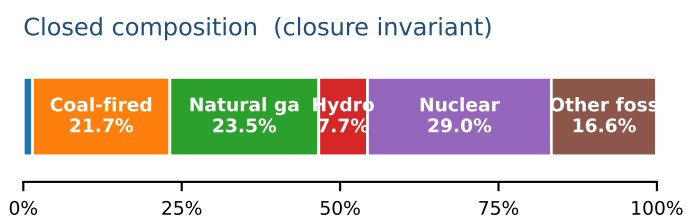




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0146	-0.462
Coal-fired	0.2166	+2.232
Natural gas	0.2351	+2.313
Hydro	0.0768	+1.195
Nuclear	0.2902	+2.524
Other fossil f	0.1662	+1.967
Solar	0.0003	-4.320
Wind	0.0001	-5.449





closure: Σ = 1.000000 OK

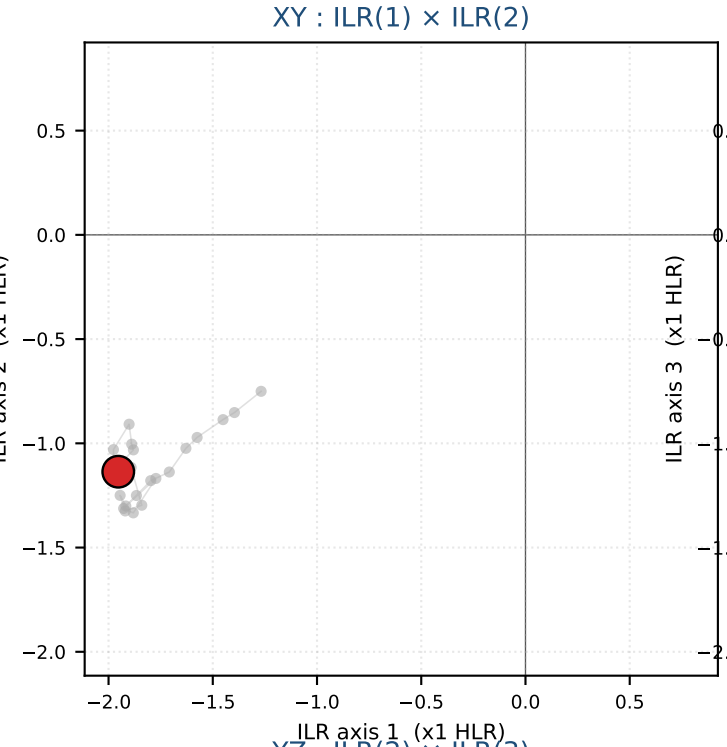
display factor: x1 (raw HLR x 1)

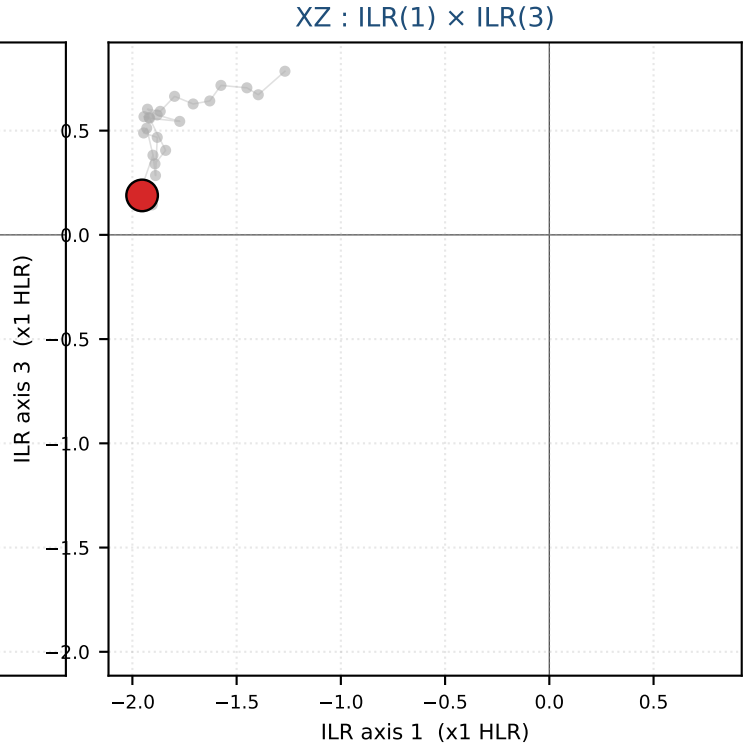
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=1 | 2001 | D=8 T=26 | level 1 first-principles

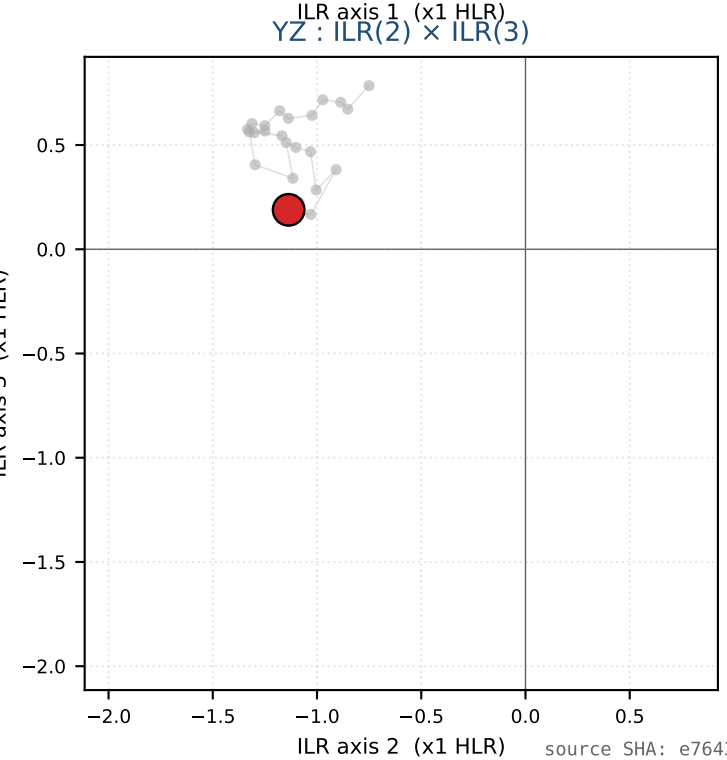
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



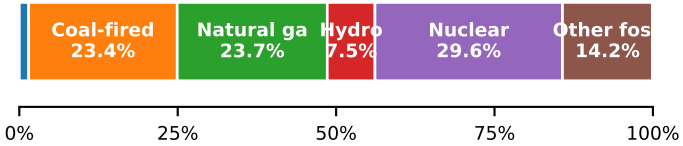


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0148	-0.599
Coal-fired	0.2344	+2.164
Natural gas	0.2368	+2.174
Hydro	0.0753	+1.028
Nuclear	0.2959	+2.397
Other fossil f	0.1420	+1.663
Solar	0.0005	-4.066
Wind	0.0002	-4.760



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

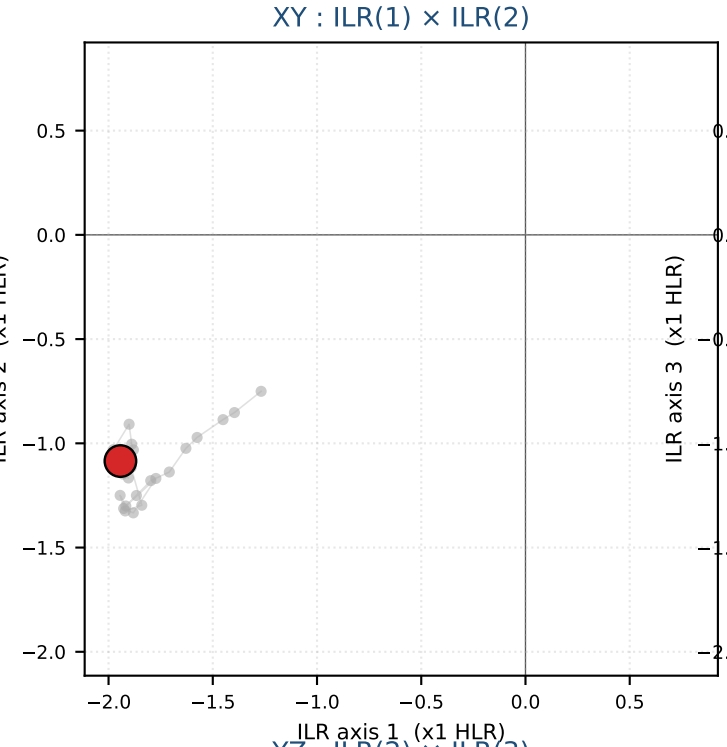
display factor: x1 (raw HLR x 1)

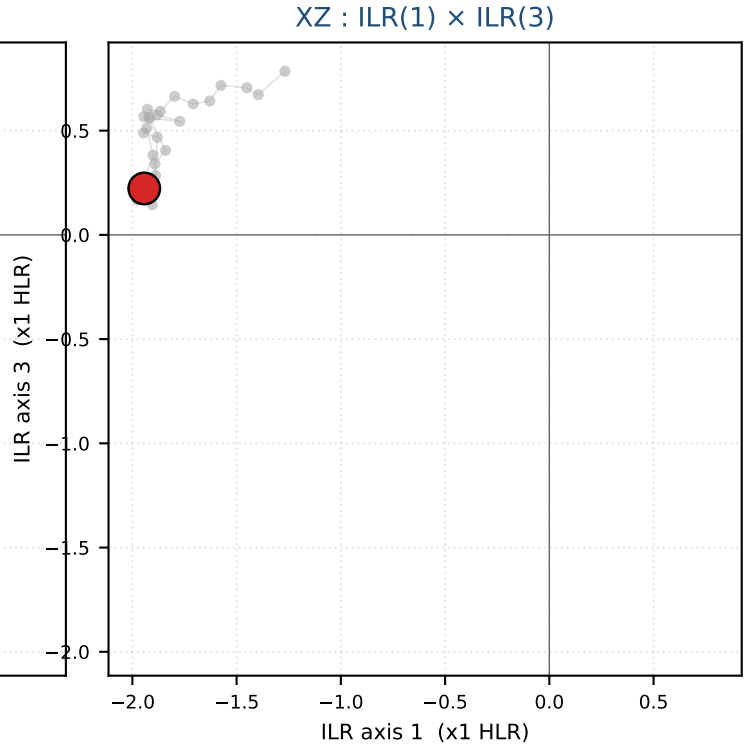
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=2 | 2002 | D=8 T=26 | level 1 first-principles

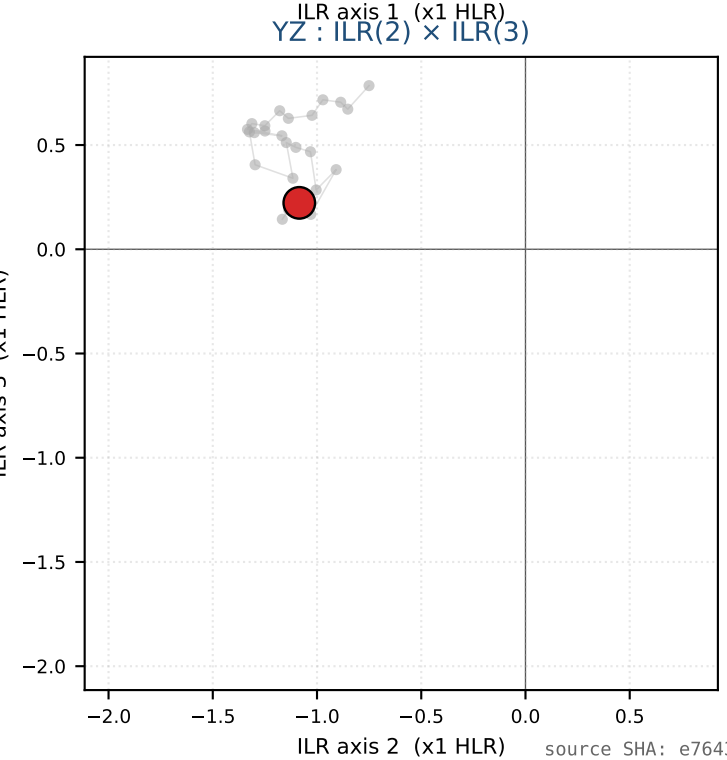
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



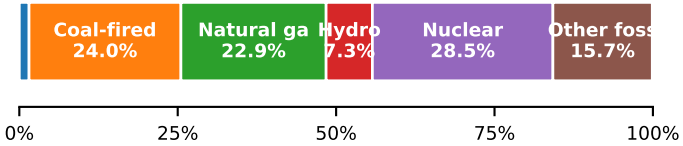


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0154	-0.667
Coal-fired	0.2397	+2.081
Natural gas	0.2291	+2.036
Hydro	0.0731	+0.893
Nuclear	0.2849	+2.254
Other fossil f	0.1568	+1.657
Solar	0.0006	-3.867
Wind	0.0004	-4.388



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

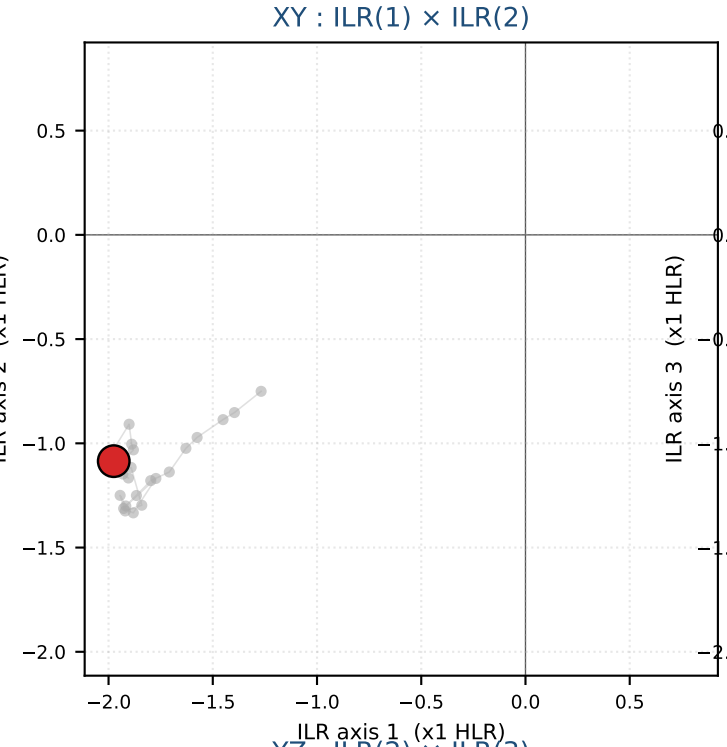
display factor: x1 (raw HLR x 1)

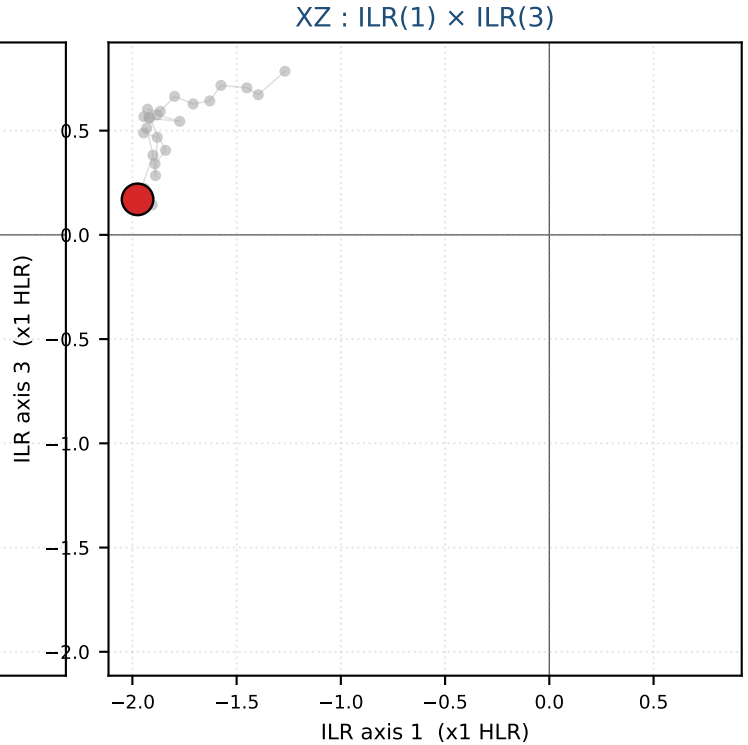
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=3 | 2003 | D=8 T=26 | level 1 first-principles

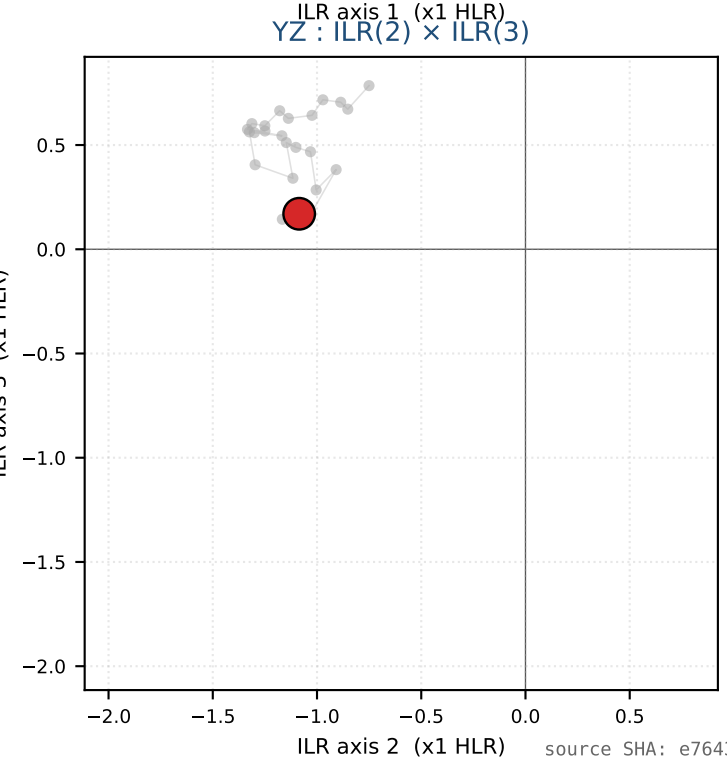
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



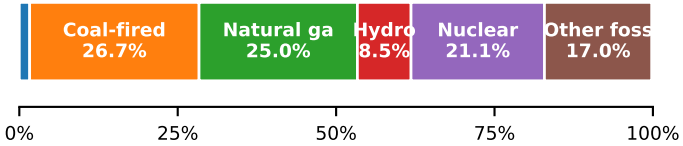


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0164	-0.756
Coal-fired	0.2673	+2.037
Natural gas	0.2500	+1.970
Hydro	0.0846	+0.887
Nuclear	0.2106	+1.798
Other fossil f	0.1695	+1.582
Solar	0.0009	-3.691
Wind	0.0008	-3.826



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

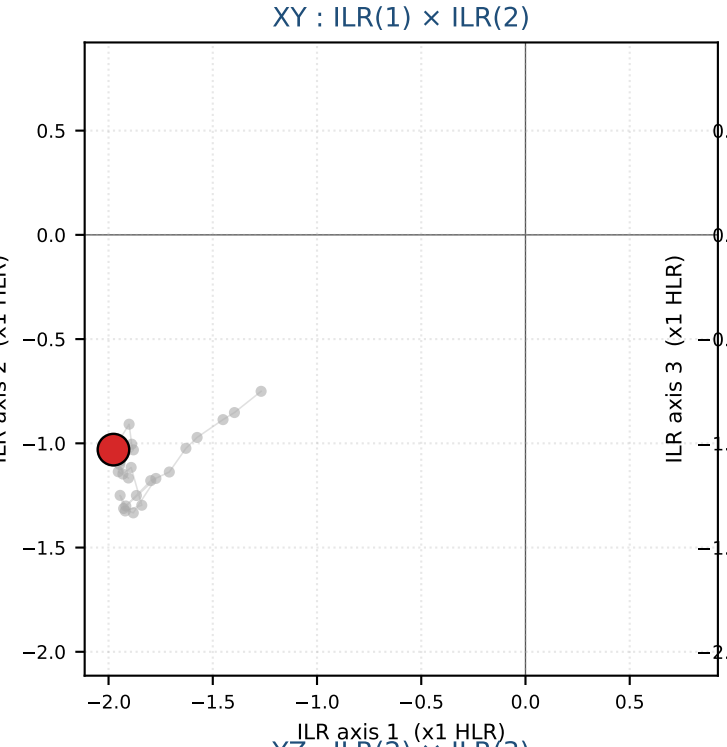
display factor: x1 (raw HLR x 1)

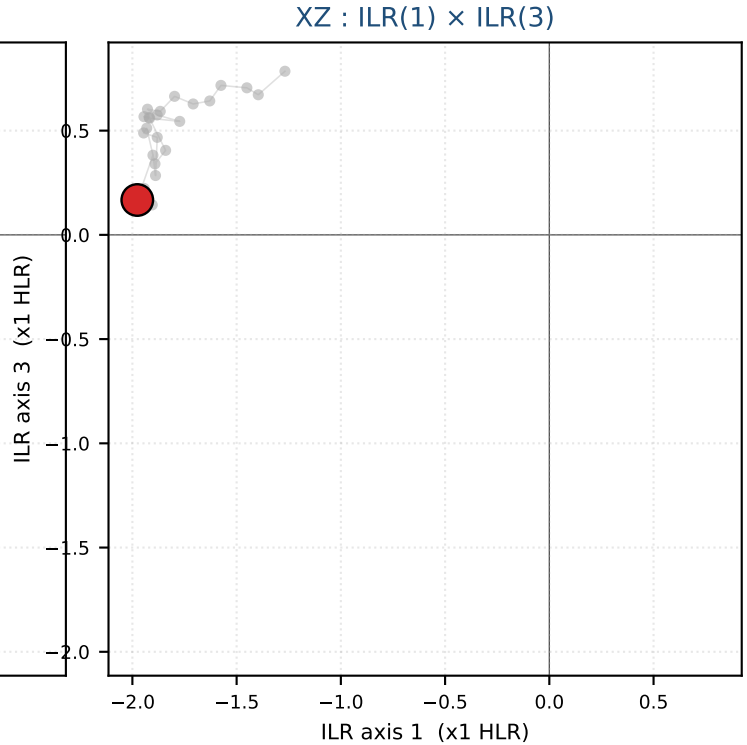
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=4 | 2004 | D=8 T=26 | level 1 first-principles

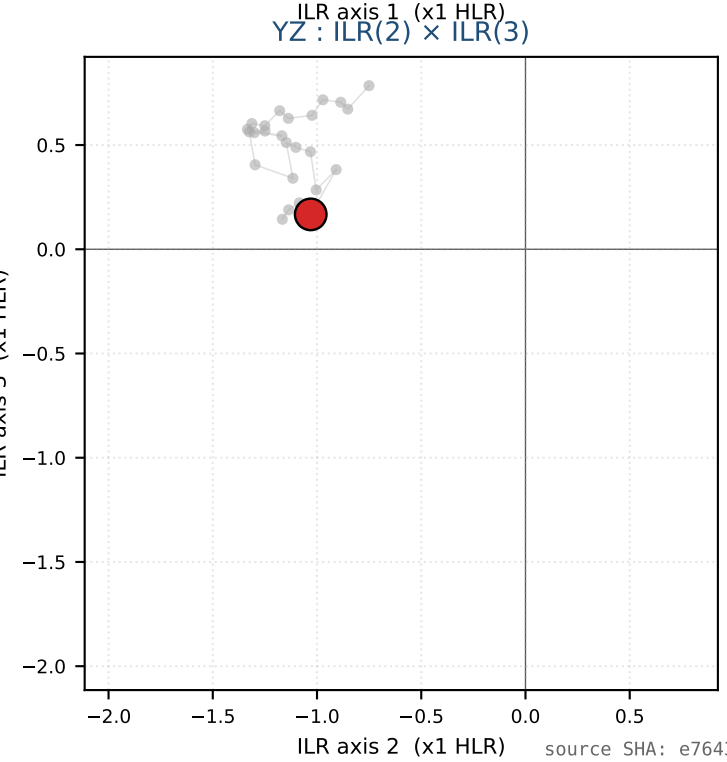
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



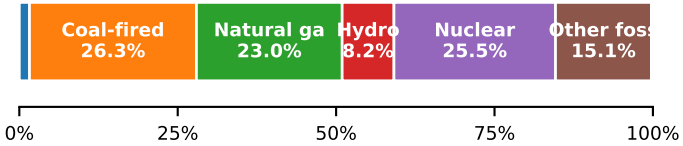


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0161	-0.862
Coal-fired	0.2635	+1.933
Natural gas	0.2300	+1.797
Hydro	0.0818	+0.763
Nuclear	0.2550	+1.900
Other fossil f	0.1513	+1.378
Solar	0.0011	-3.517
Wind	0.0013	-3.391



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

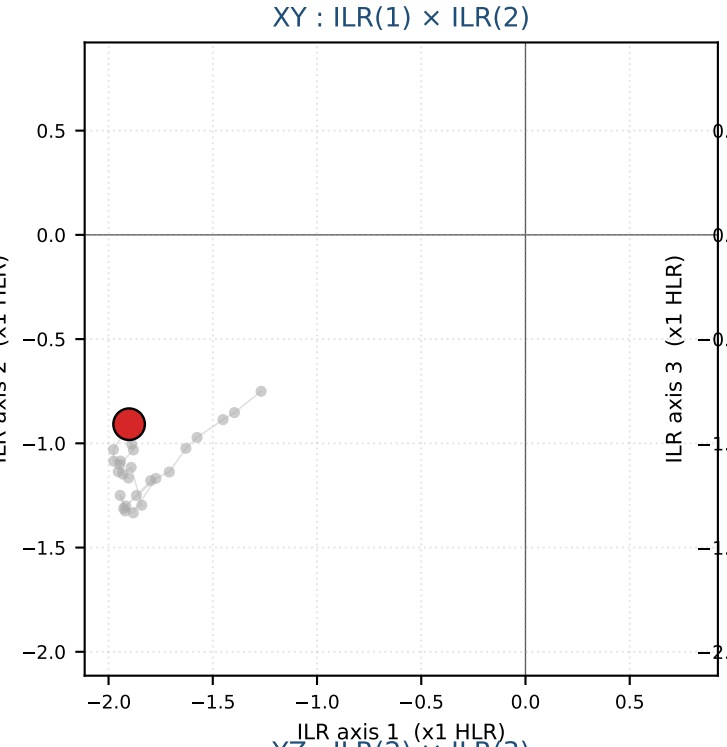
display factor: x1 (raw HLR x 1)

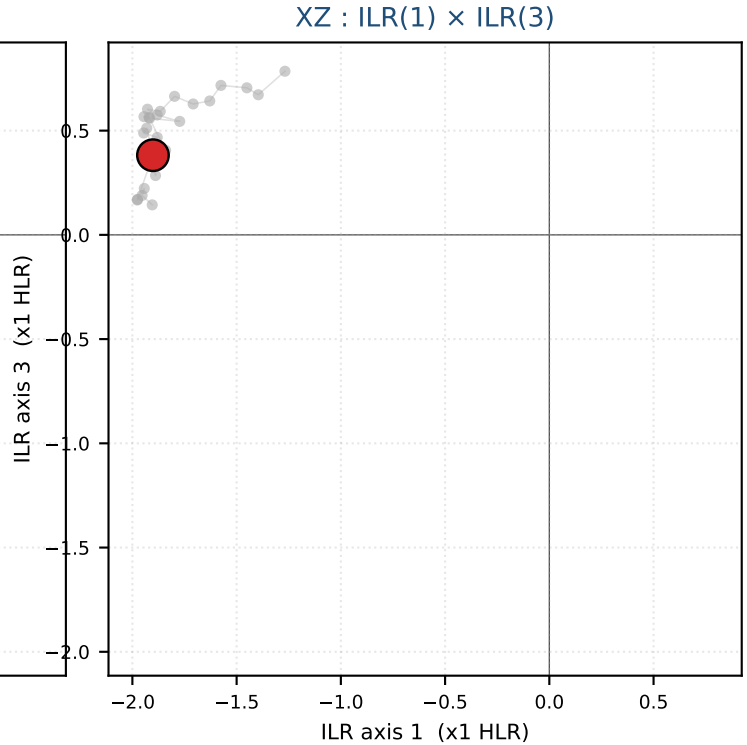
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=5 | 2005 | D=8 T=26 | level 1 first-principles

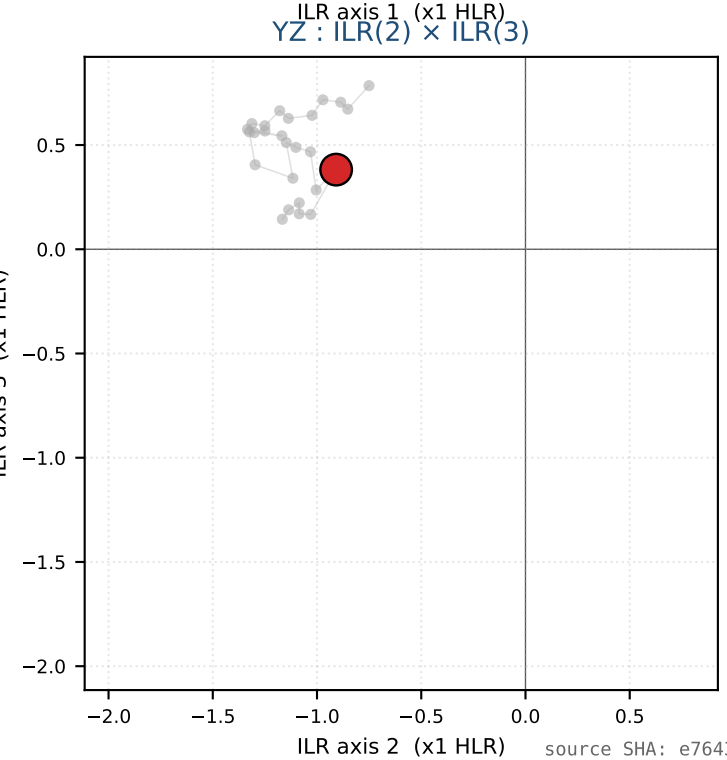
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

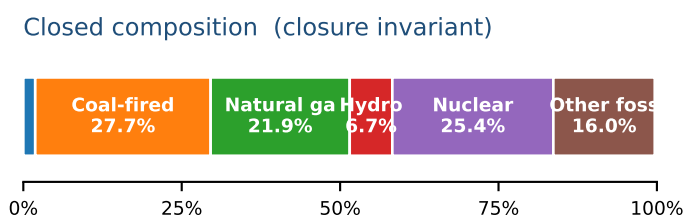




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0188	-0.769
Coal-fired	0.2768	+1.920
Natural gas	0.2195	+1.688
Hydro	0.0673	+0.506
Nuclear	0.2541	+1.835
Other fossil f	0.1604	+1.375
Solar	0.0014	-3.357
Wind	0.0017	-3.198





closure: Σ = 1.000000 OK

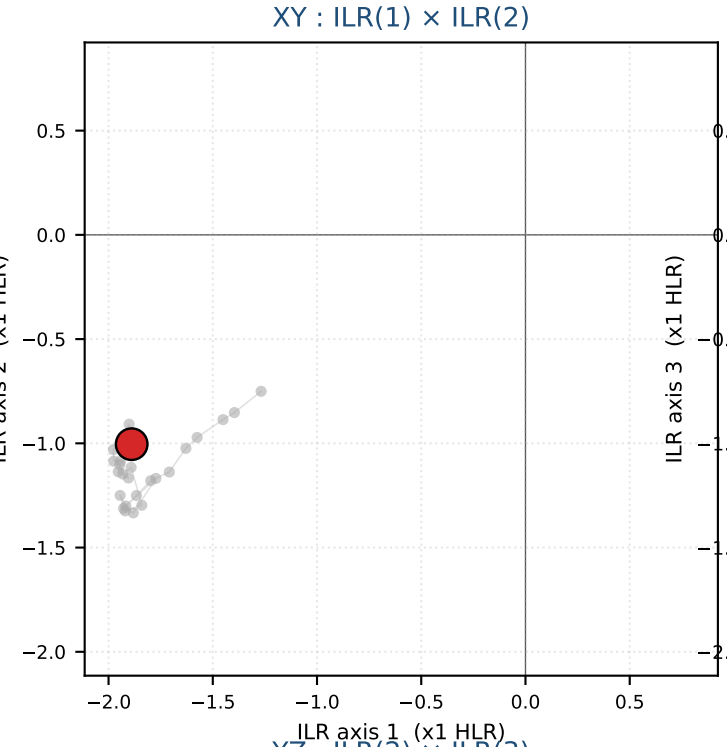
display factor: x1 (raw HLR x 1)

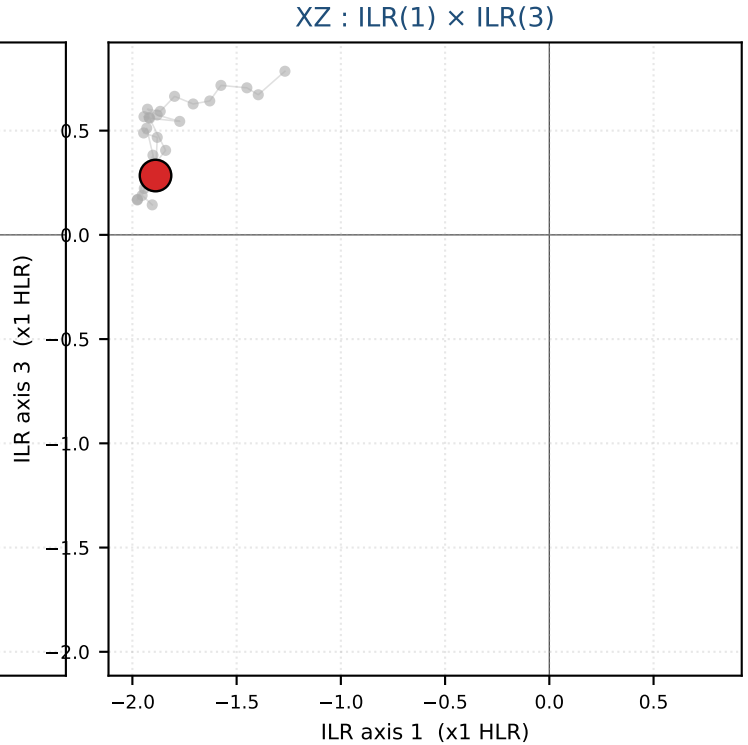
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=6 | 2006 | D=8 T=26 | level 1 first-principles

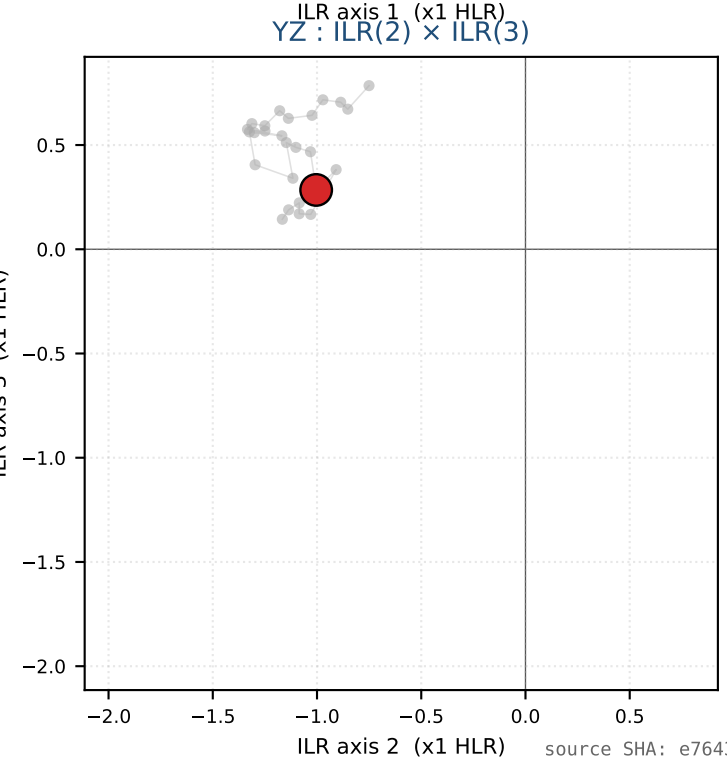
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



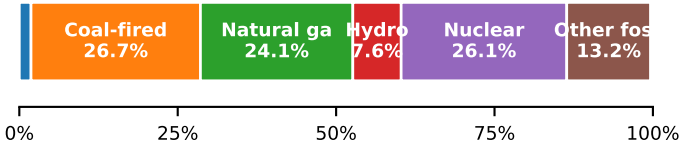


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0185	-0.823
Coal-fired	0.2674	+1.849
Natural gas	0.2405	+1.743
Hydro	0.0763	+0.594
Nuclear	0.2613	+1.826
Other fossil f	0.1324	+1.146
Solar	0.0017	-3.199
Wind	0.0018	-3.136



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

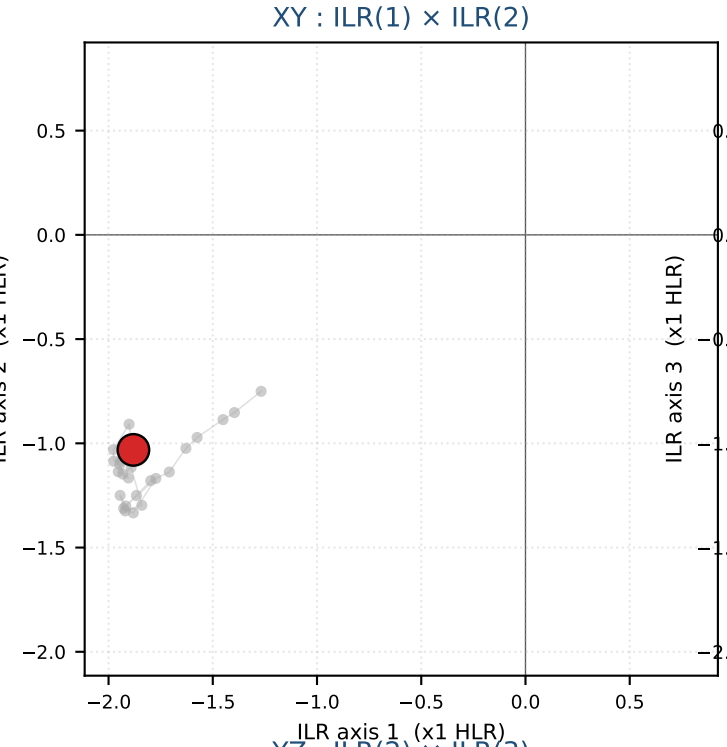
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

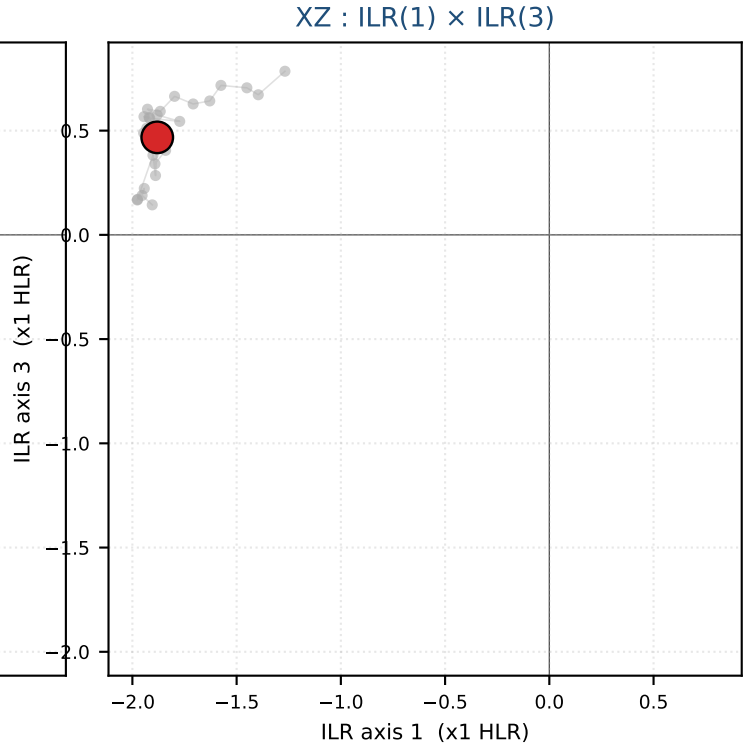
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=7 | 2007 | D=8 T=26 | level 1 first-principles

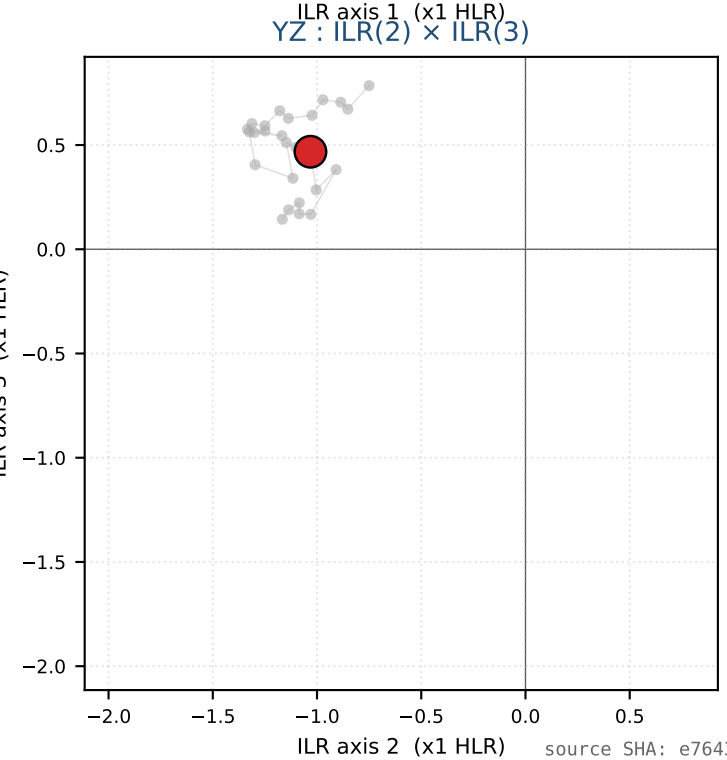
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

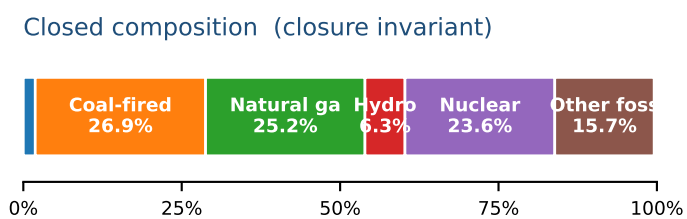




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0188	-0.845
Coal-fired	0.2689	+1.815
Natural gas	0.2515	+1.748
Hydro	0.0631	+0.366
Nuclear	0.2364	+1.686
Other fossil f	0.1569	+1.276
Solar	0.0020	-3.108
Wind	0.0023	-2.937





closure: Σ = 1.000000 OK

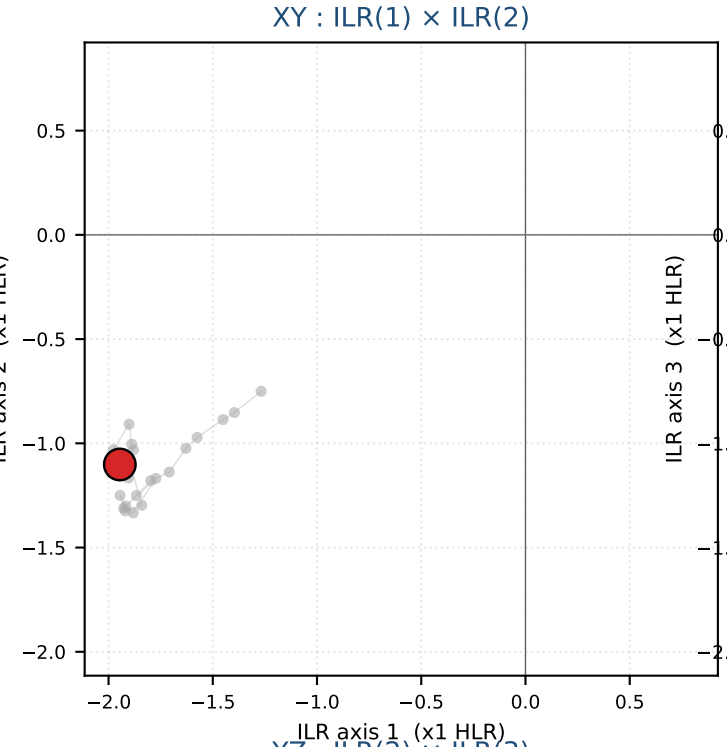
display factor: x1 (raw HLR x 1)

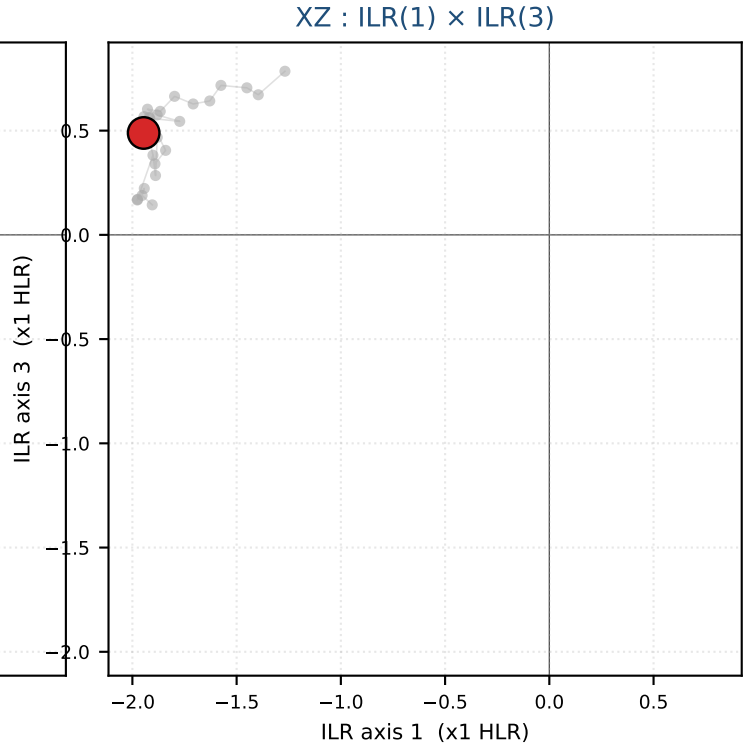
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=8 | 2008 | D=8 T=26 | level 1 first-principles

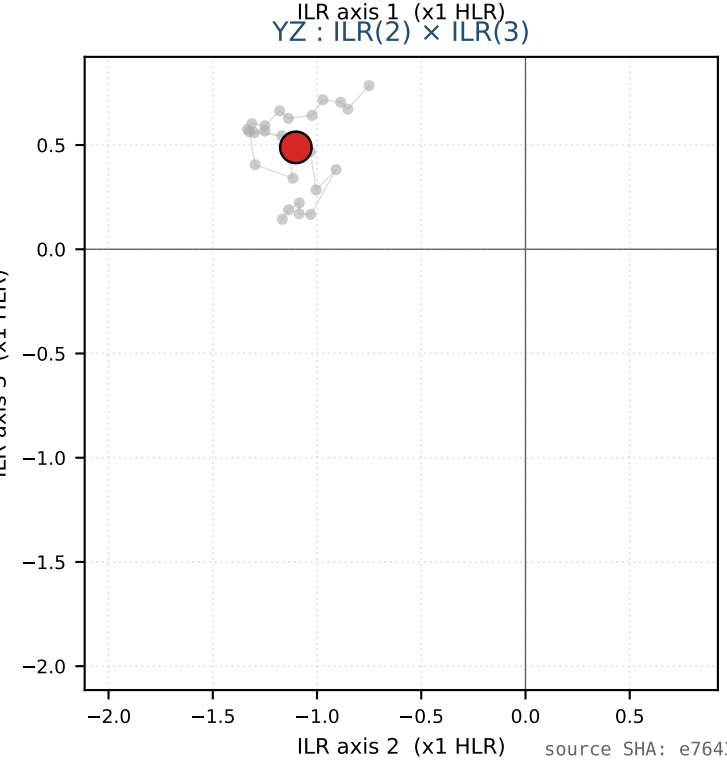
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

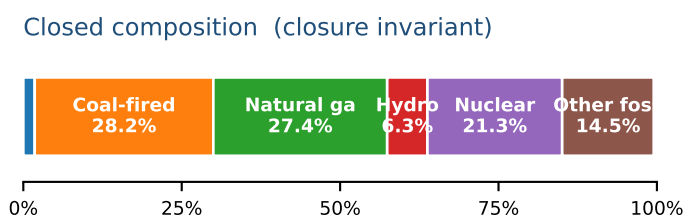




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0180	-0.901
Coal-fired	0.2819	+1.851
Natural gas	0.2744	+1.824
Hydro	0.0635	+0.360
Nuclear	0.2127	+1.569
Other fossil f	0.1448	+1.185
Solar	0.0022	-3.008
Wind	0.0025	-2.881





closure: Σ = 1.000000 OK

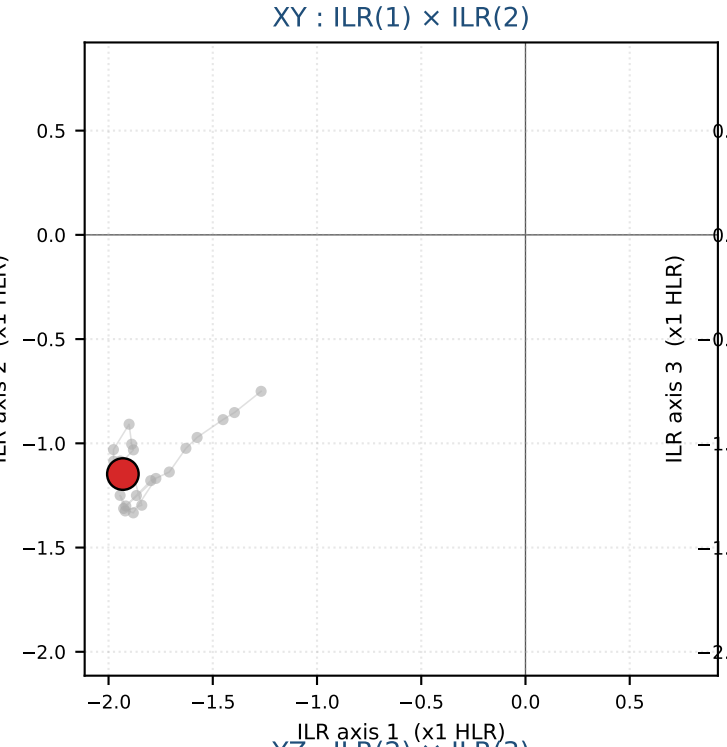
display factor: x1 (raw HLR x 1)

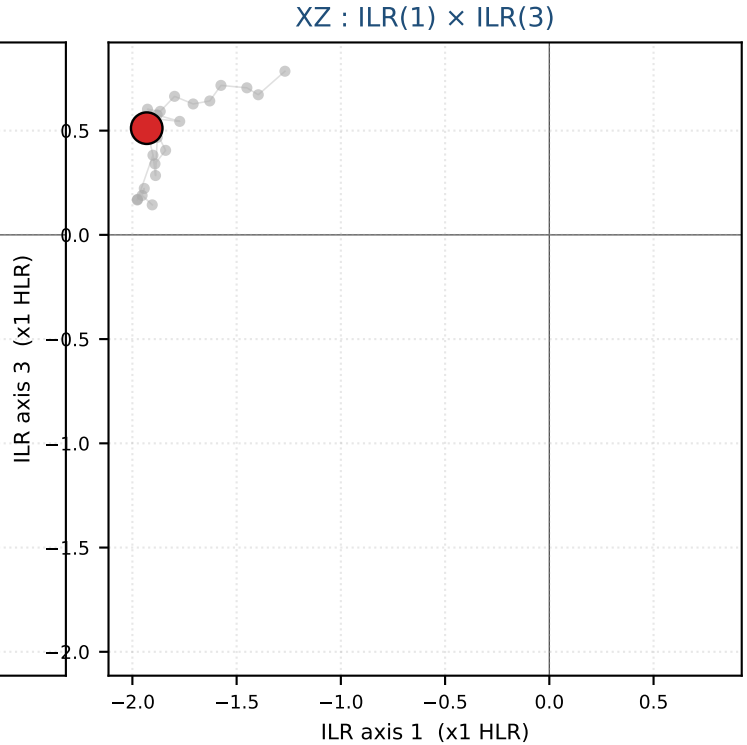
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=9 | 2009 | D=8 T=26 | level 1 first-principles

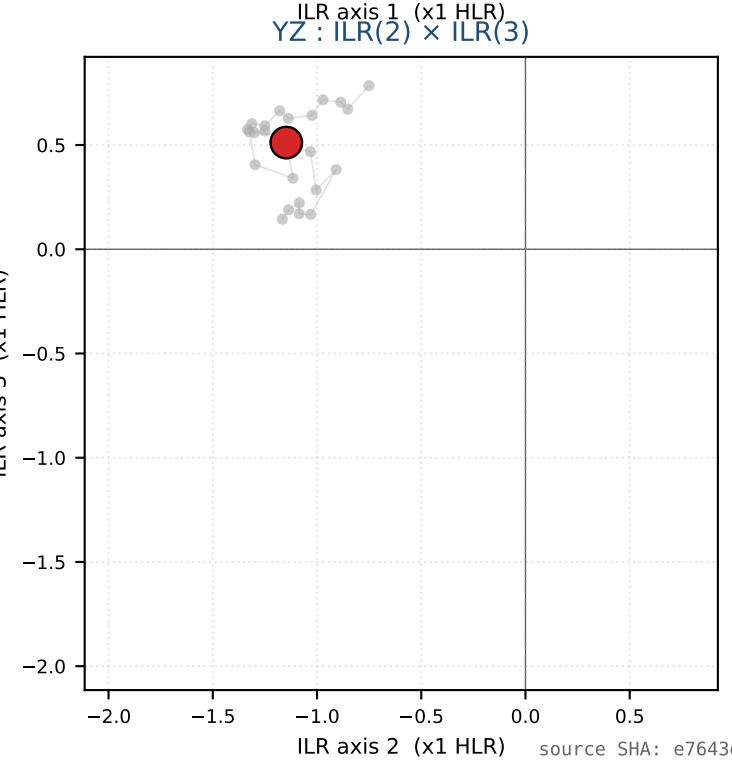
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

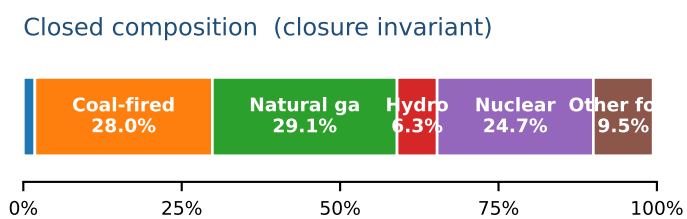




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0182	-0.915
Coal-fired	0.2801	+1.817
Natural gas	0.2915	+1.856
Hydro	0.0633	+0.329
Nuclear	0.2465	+1.689
Other fossil f	0.0945	+0.730
Solar	0.0027	-2.812
Wind	0.0031	-2.694





closure: Σ = 1.000000 OK

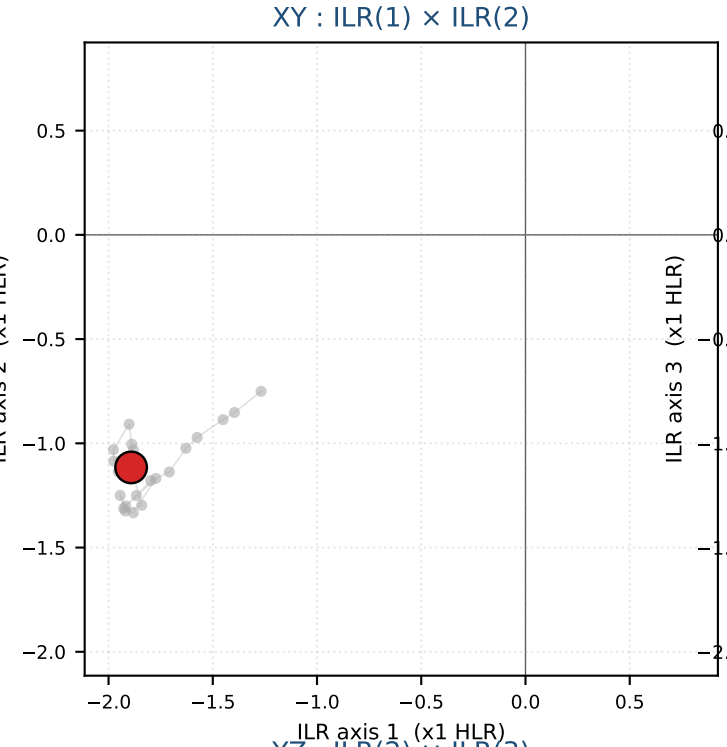
display factor: x1 (raw HLR x 1)

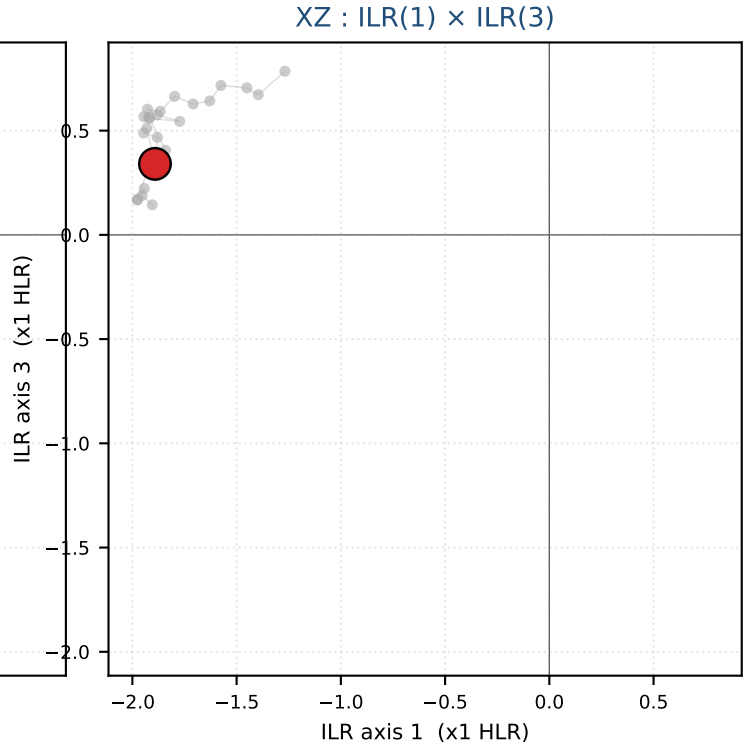
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=10 | 2010 | D=8 T=26 | level 1 first-principles

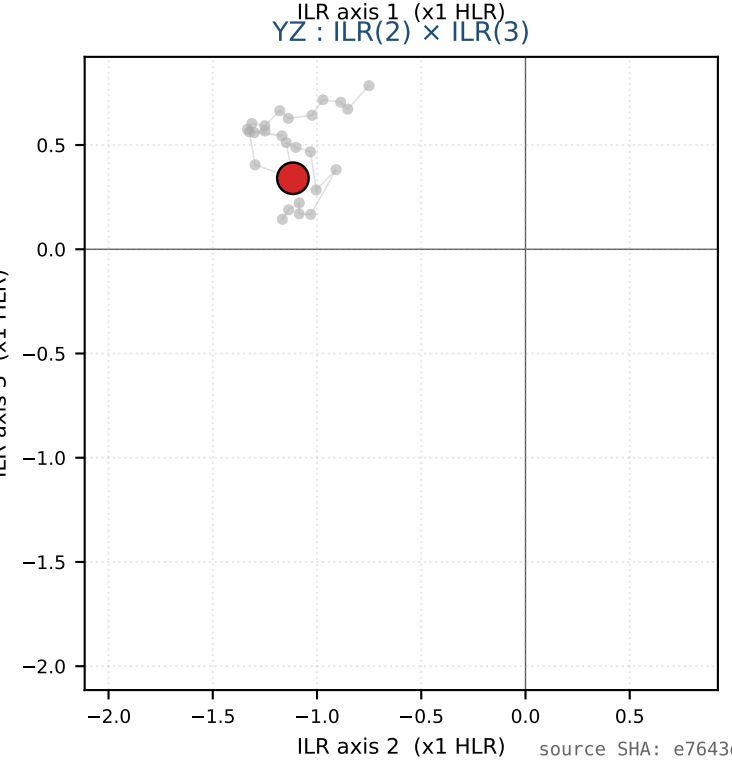
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

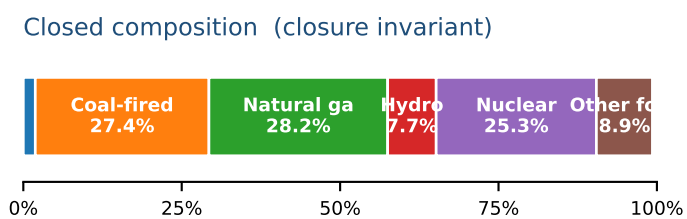




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0189	-0.938
Coal-fired	0.2741	+1.737
Natural gas	0.2821	+1.766
Hydro	0.0766	+0.462
Nuclear	0.2529	+1.657
Other fossil f	0.0887	+0.609
Solar	0.0034	-2.640
Wind	0.0034	-2.653





closure: Σ = 1.000000 OK

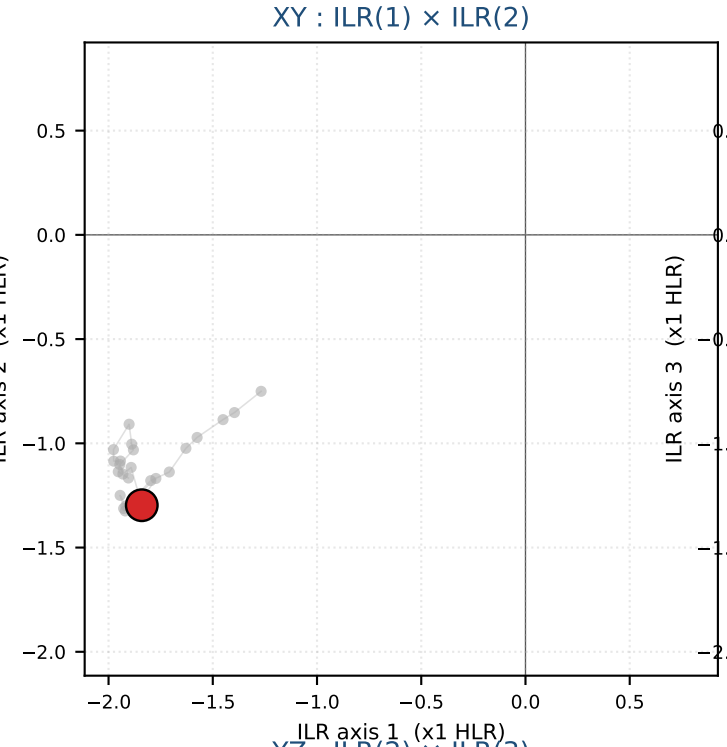
display factor: x1 (raw HLR x 1)

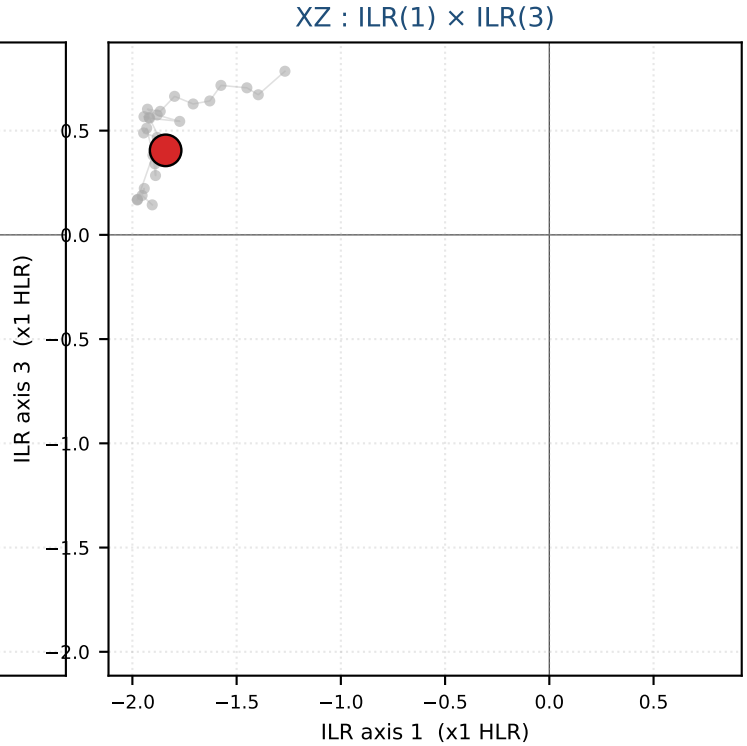
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=11 | 2011 | D=8 T=26 | level 1 first-principles

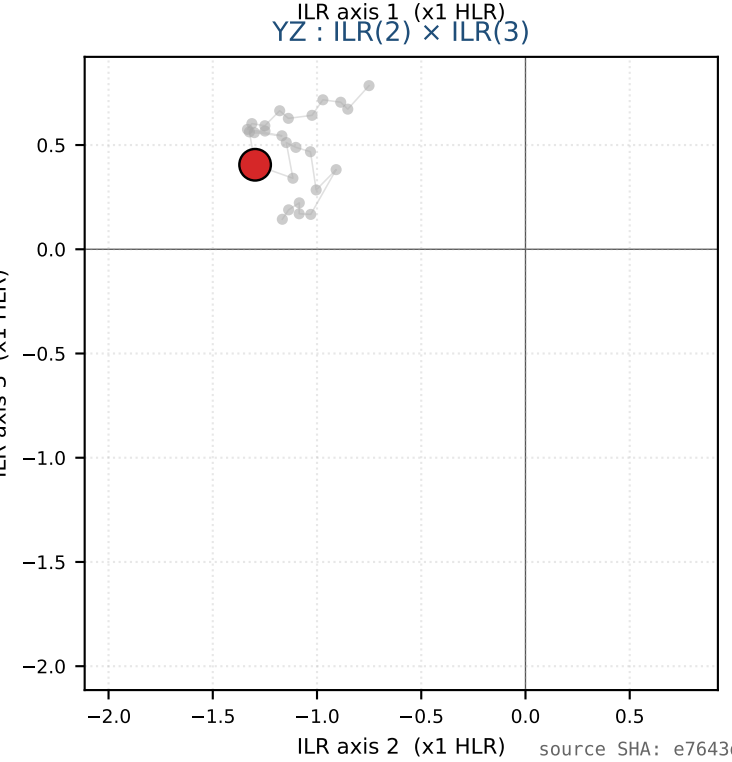
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

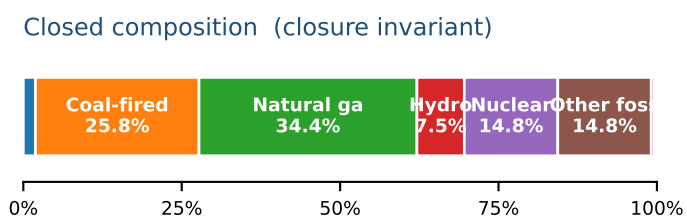




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0191	-1.004
Coal-fired	0.2582	+1.599
Natural gas	0.3440	+1.886
Hydro	0.0747	+0.359
Nuclear	0.1476	+1.040
Other fossil f	0.1475	+1.039
Solar	0.0049	-2.360
Wind	0.0040	-2.559





closure: Σ = 1.000000 OK

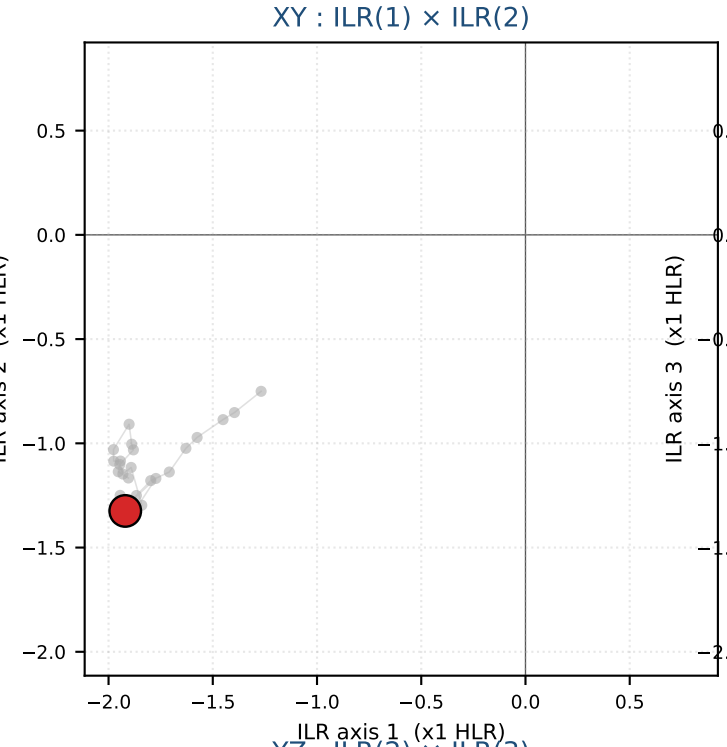
display factor: x1 (raw HLR x 1)

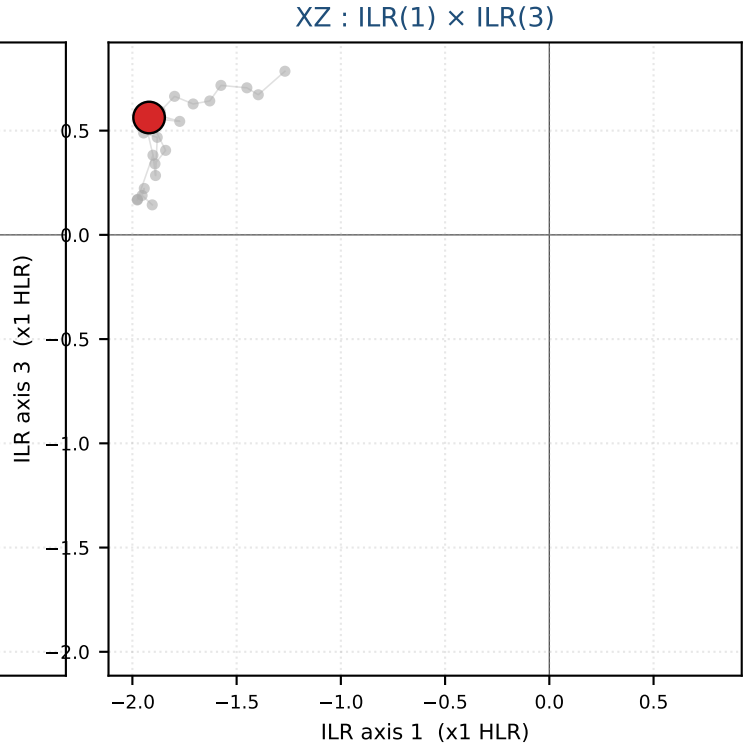
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=12 | 2012 | D=8 T=26 | level 1 first-principles

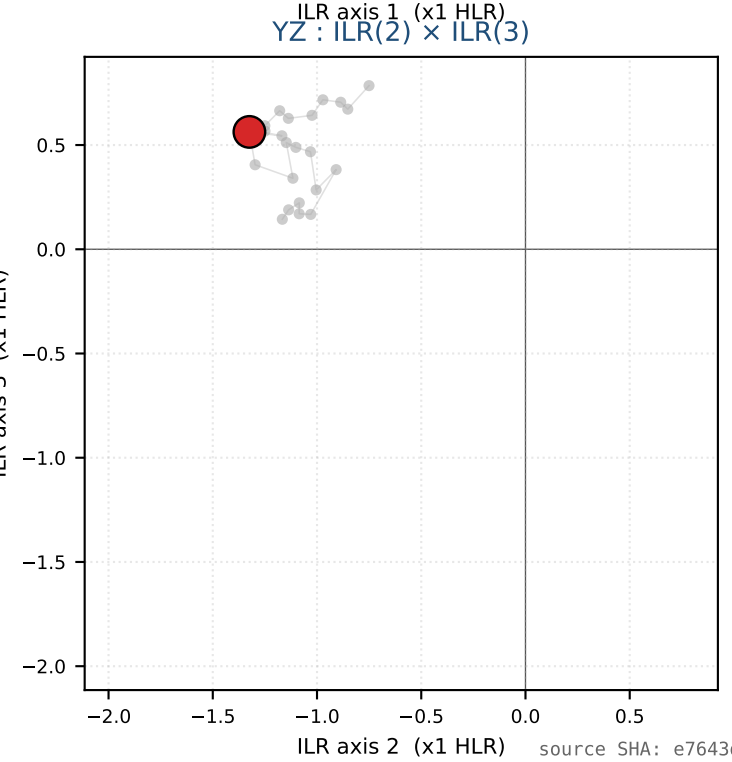
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



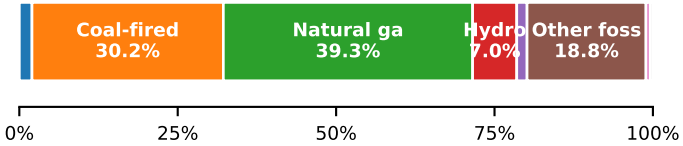


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0200	-0.792
Coal-fired	0.3020	+1.924
Natural gas	0.3934	+2.188
Hydro	0.0697	+0.457
Nuclear	0.0163	-0.999
Other fossil f	0.1877	+1.448
Solar	0.0067	-1.891
Wind	0.0043	-2.335



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

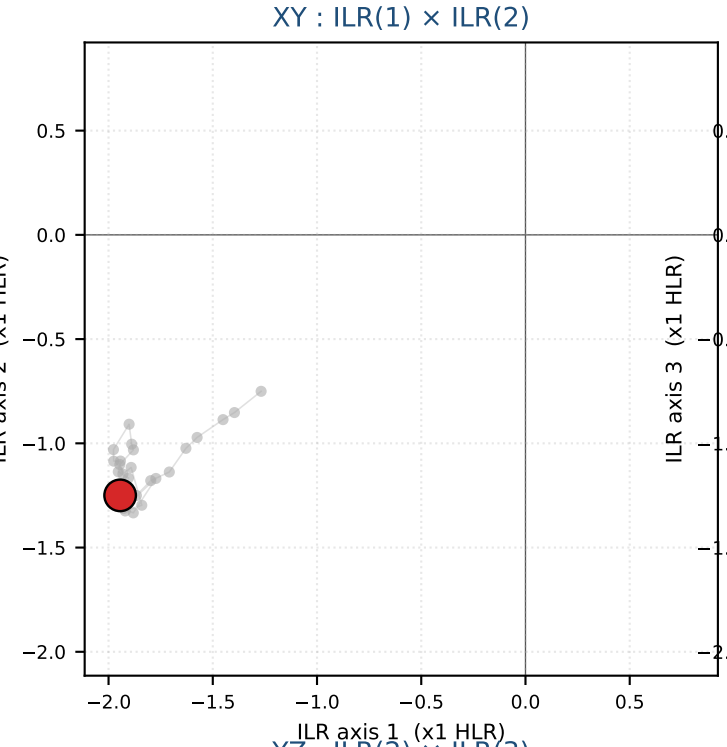
display factor: x1 (raw HLR x 1)

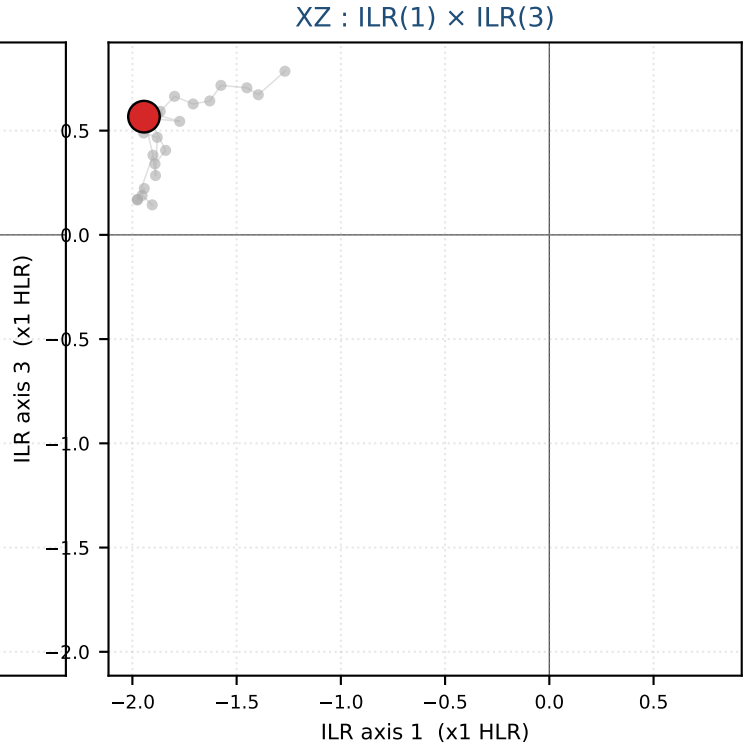
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=13 | 2013 | D=8 T=26 | level 1 first-principles

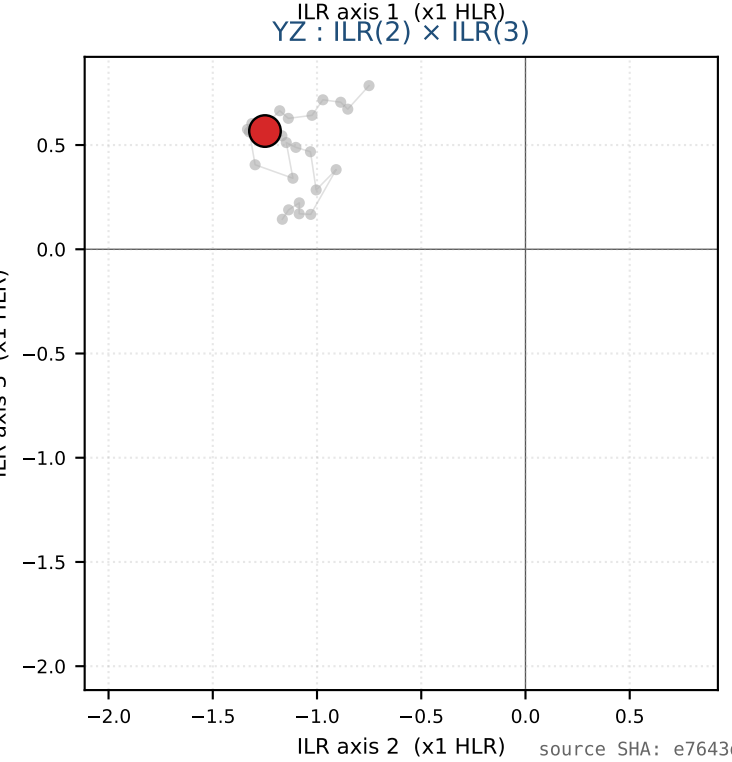
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



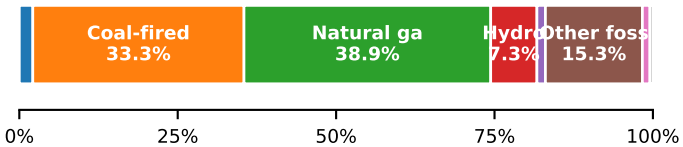


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0213	-0.788
Coal-fired	0.3333	+1.962
Natural gas	0.3894	+2.118
Hydro	0.0729	+0.443
Nuclear	0.0134	-1.250
Other fossil f	0.1530	+1.184
Solar	0.0119	-1.373
Wind	0.0047	-2.296



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

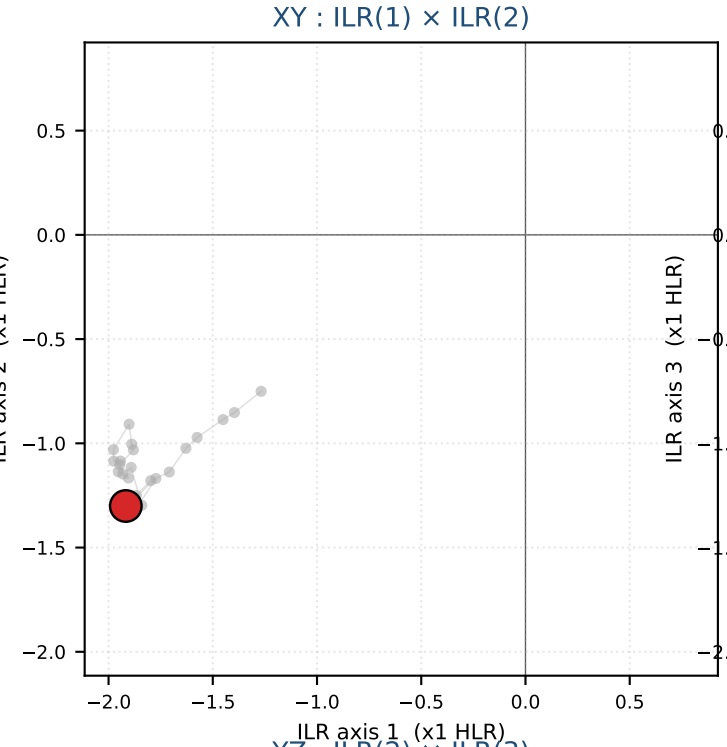
display factor: x1 (raw HLR x 1)

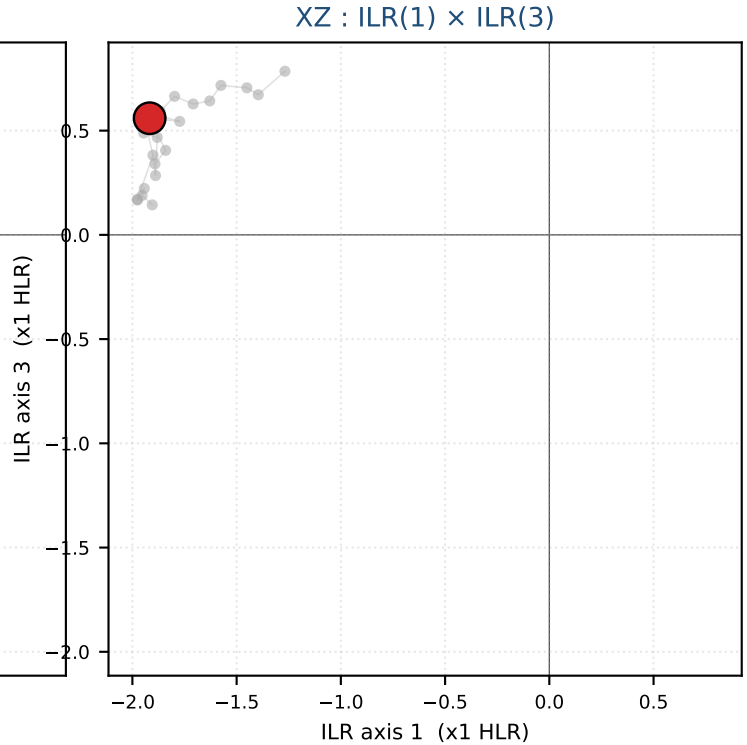
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000

ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408

ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=14 | 2014 | D=8 T=26 | level 1 first-principles

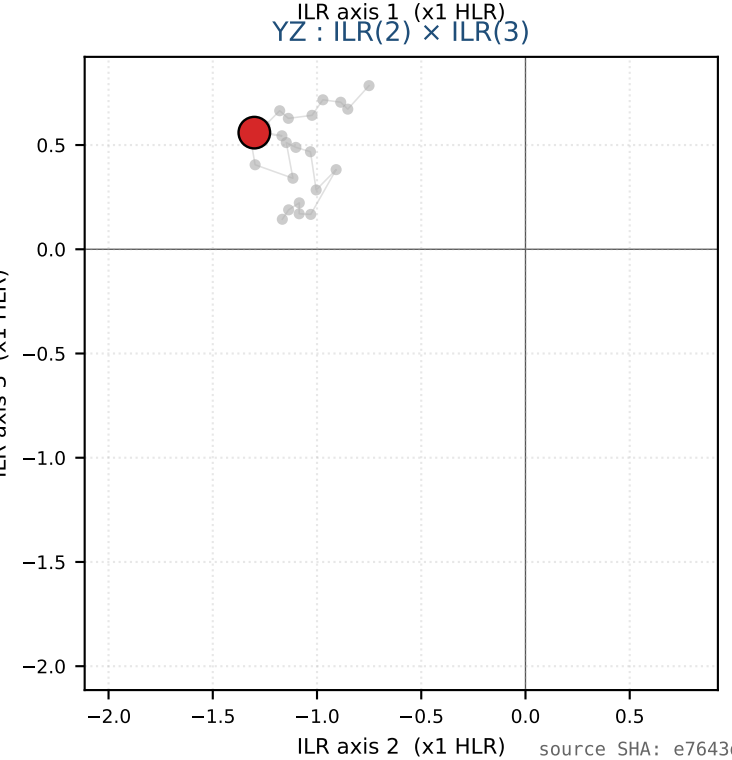
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

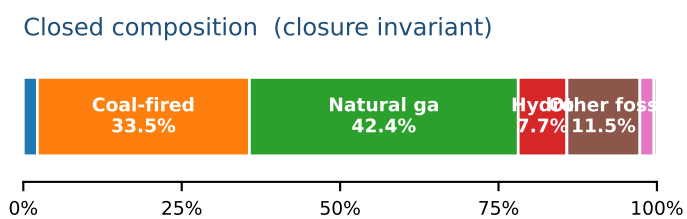




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0222	+3.839
Coal-fired	0.3346	+6.550
Natural gas	0.4243	+6.788
Hydro	0.0769	+5.080
Nuclear	0.0000	-33.862
Other fossil f	0.1151	+5.483
Solar	0.0222	+3.835
Wind	0.0047	+2.288





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

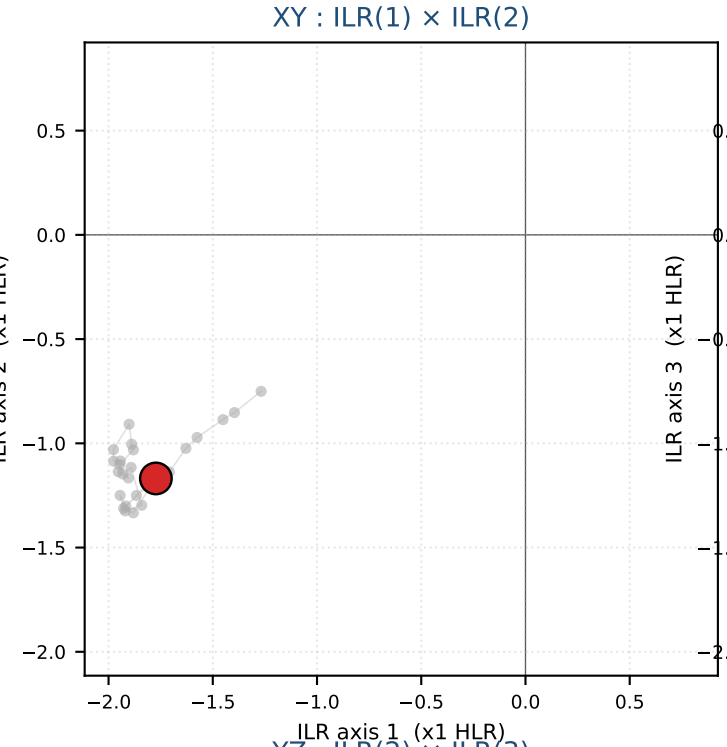
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

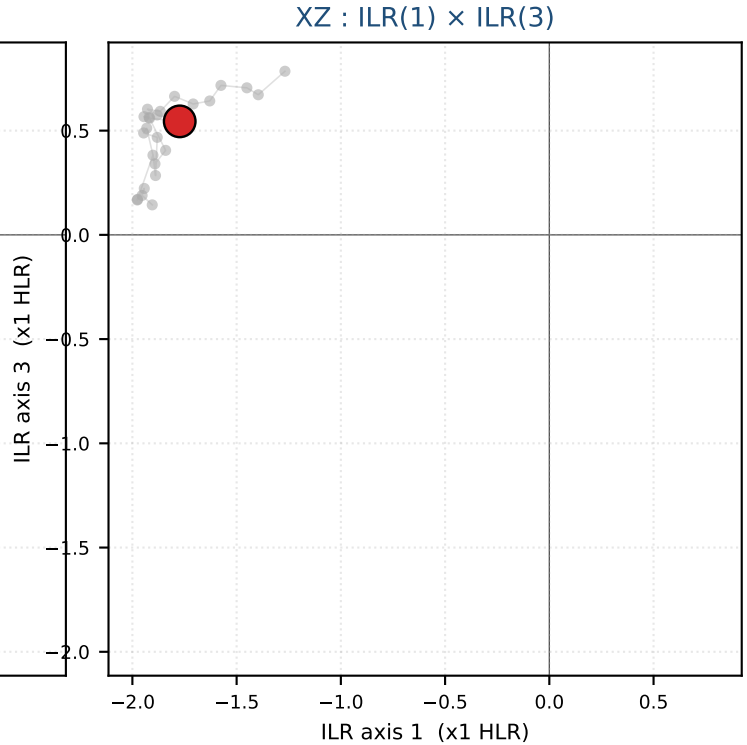
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=15 | 2015 | D=8 T=26 | level 1 first-principles

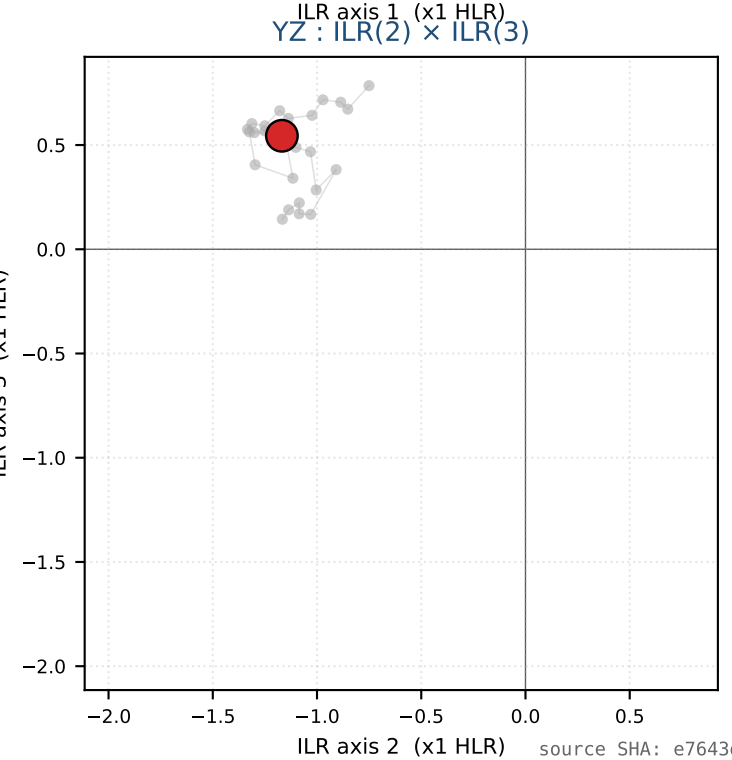
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

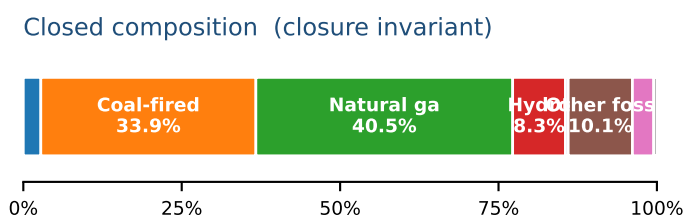




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0277	-0.531
Coal-fired	0.3394	+1.976
Natural gas	0.4053	+2.154
Hydro	0.0833	+0.571
Nuclear	0.0044	-2.372
Other fossil f	0.1014	+0.768
Solar	0.0335	-0.338
Wind	0.0051	-2.228





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

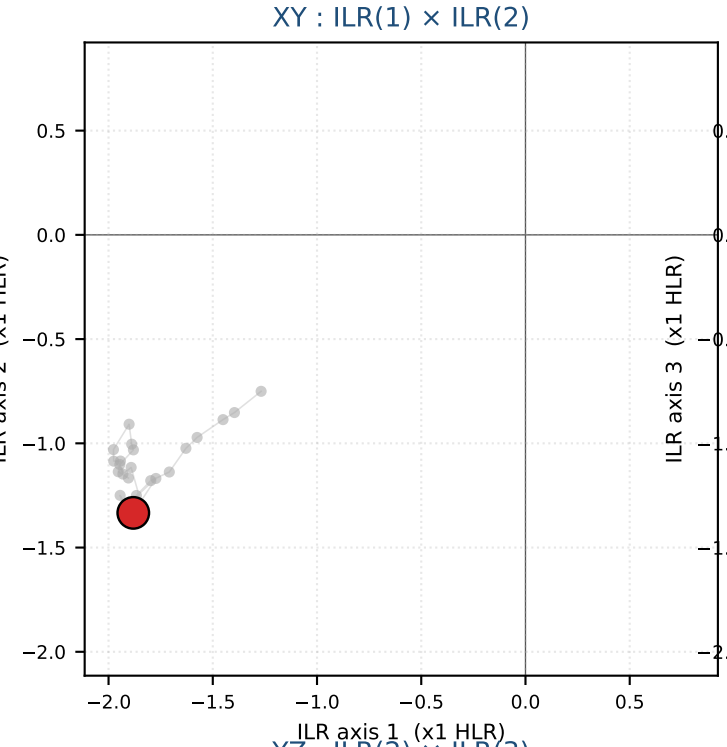
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

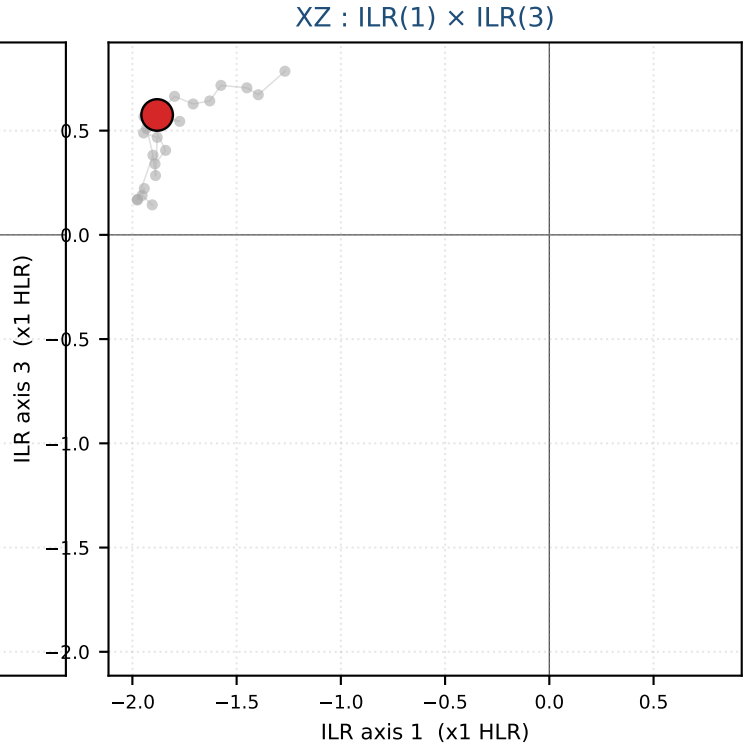
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=16 | 2016 | D=8 T=26 | level 1 first-principles

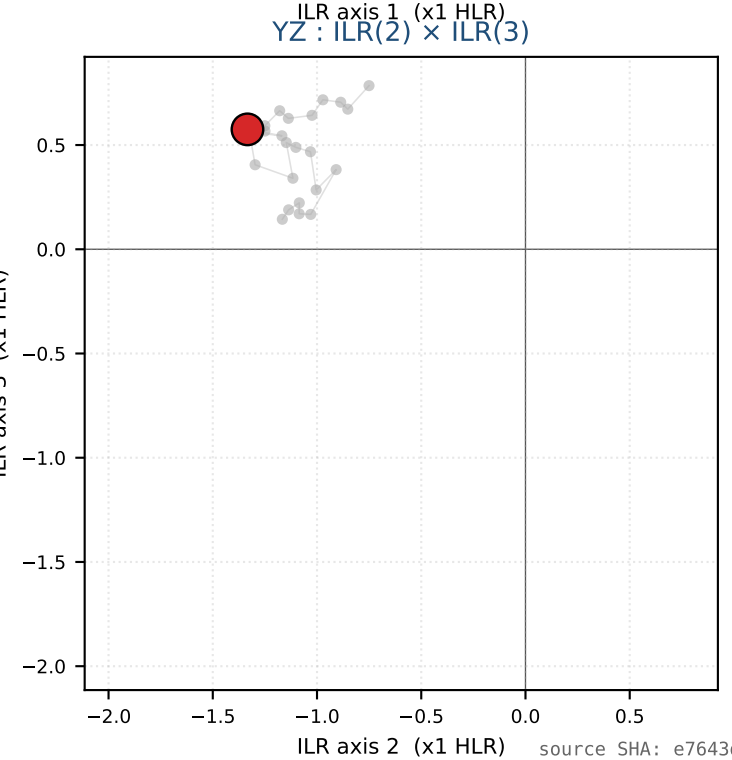
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

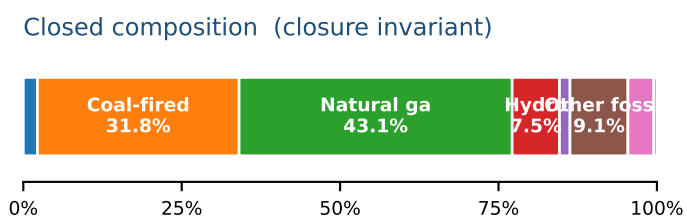




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0223	-0.884
Coal-fired	0.3183	+1.777
Natural gas	0.4310	+2.080
Hydro	0.0747	+0.327
Nuclear	0.0166	-1.176
Other fossil f	0.0914	+0.529
Solar	0.0407	-0.279
Wind	0.0050	-2.373





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

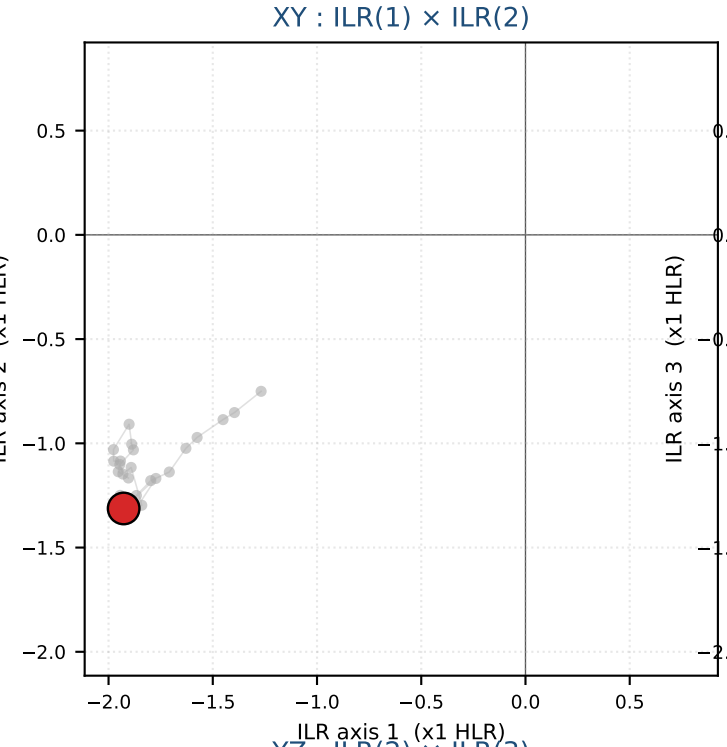
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

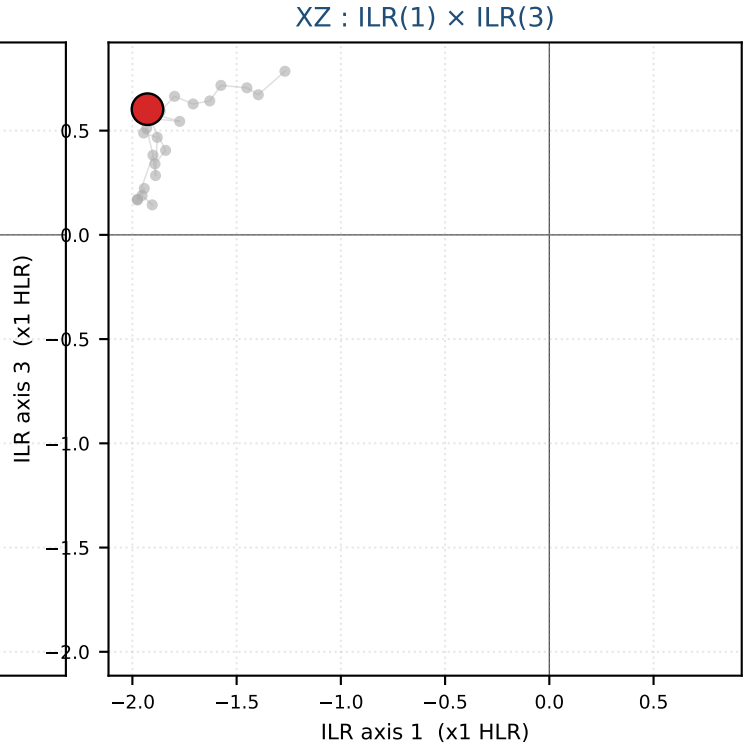
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=17 | 2017 | D=8 T=26 | level 1 first-principles

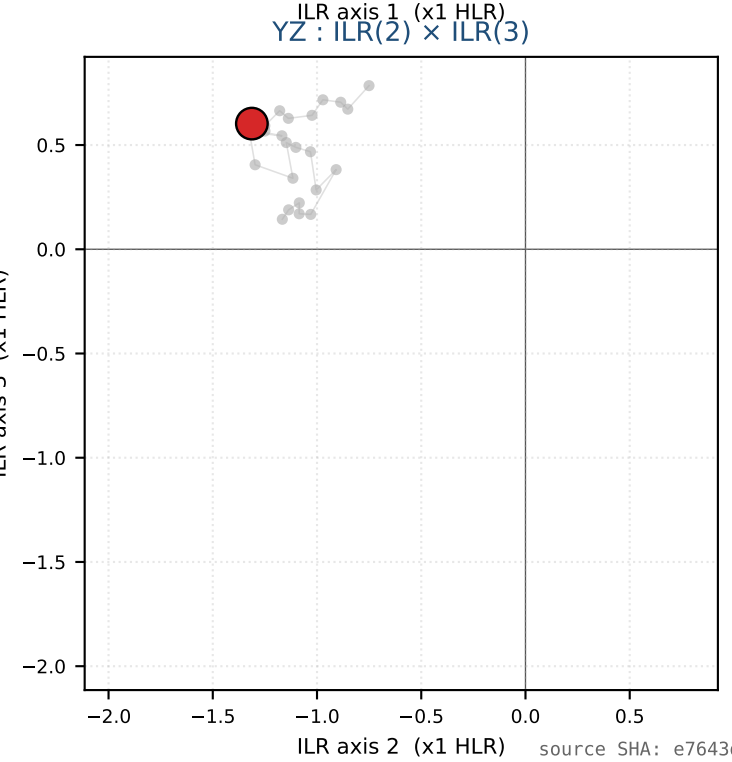
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

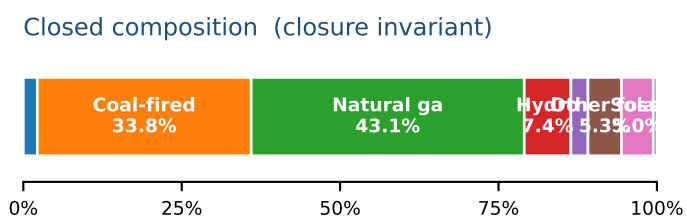




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0221	-0.931
Coal-fired	0.3375	+1.795
Natural gas	0.4310	+2.039
Hydro	0.0736	+0.272
Nuclear	0.0270	-0.731
Other fossil f	0.0528	-0.060
Solar	0.0502	-0.111
Wind	0.0058	-2.273





closure: Σ = 1.000000 OK

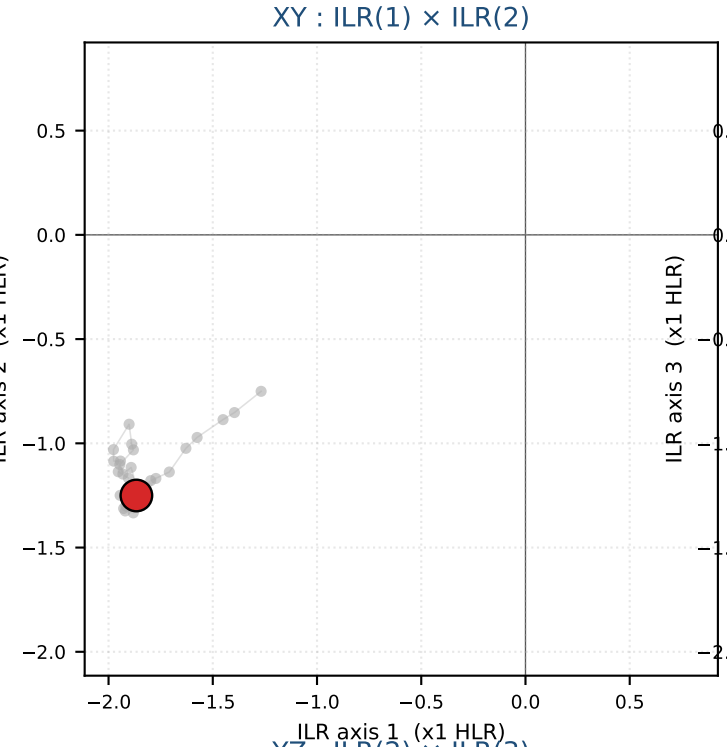
display factor: x1 (raw HLR x 1)

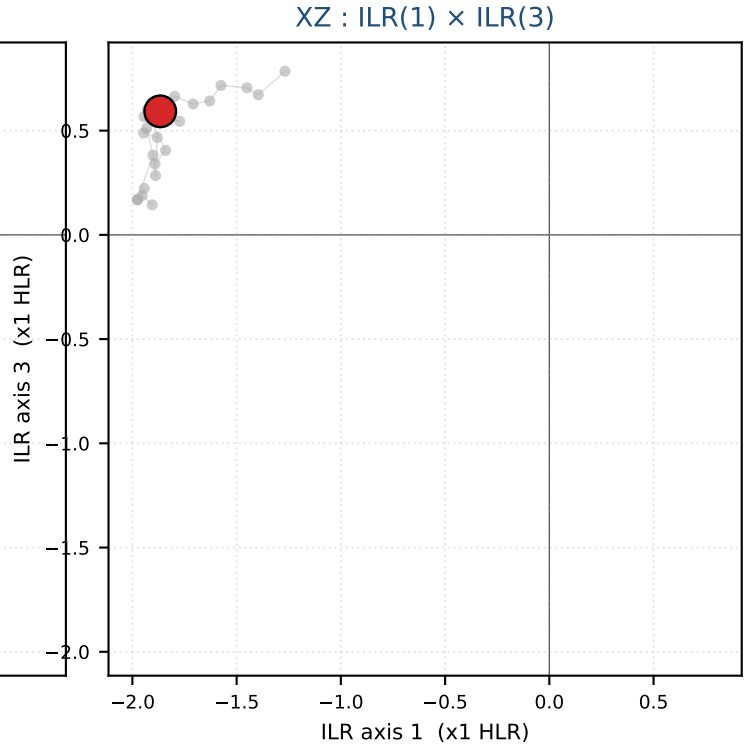
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=18 | 2018 | D=8 T=26 | level 1 first-principles

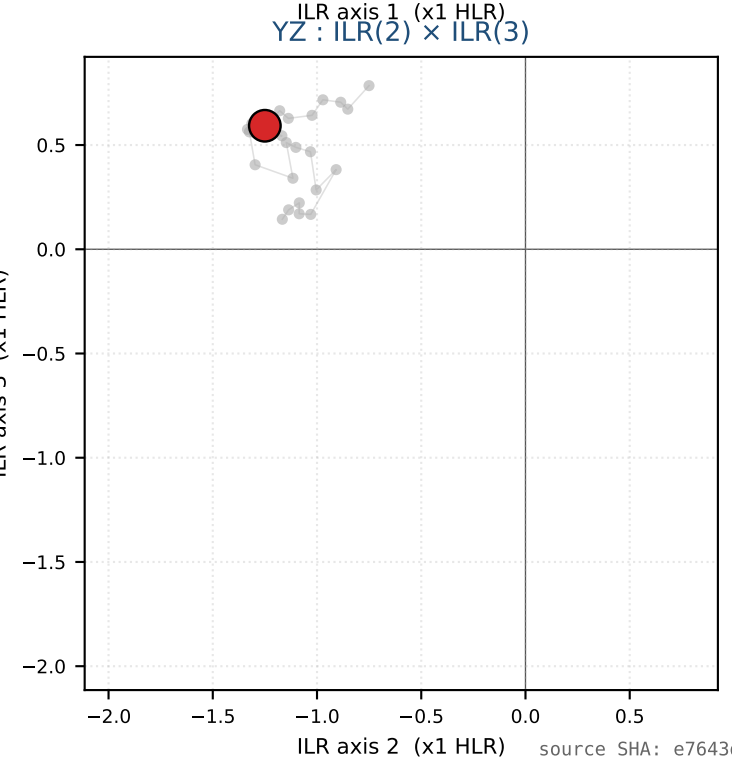
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



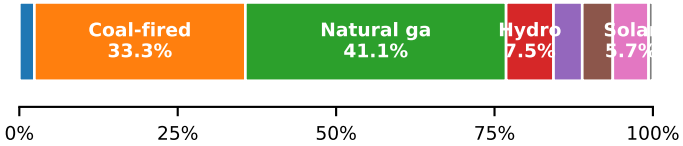


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0238	-0.951
Coal-fired	0.3330	+1.689
Natural gas	0.4115	+1.901
Hydro	0.0748	+0.196
Nuclear	0.0453	-0.305
Other fossil f	0.0482	-0.244
Solar	0.0565	-0.084
Wind	0.0068	-2.202



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

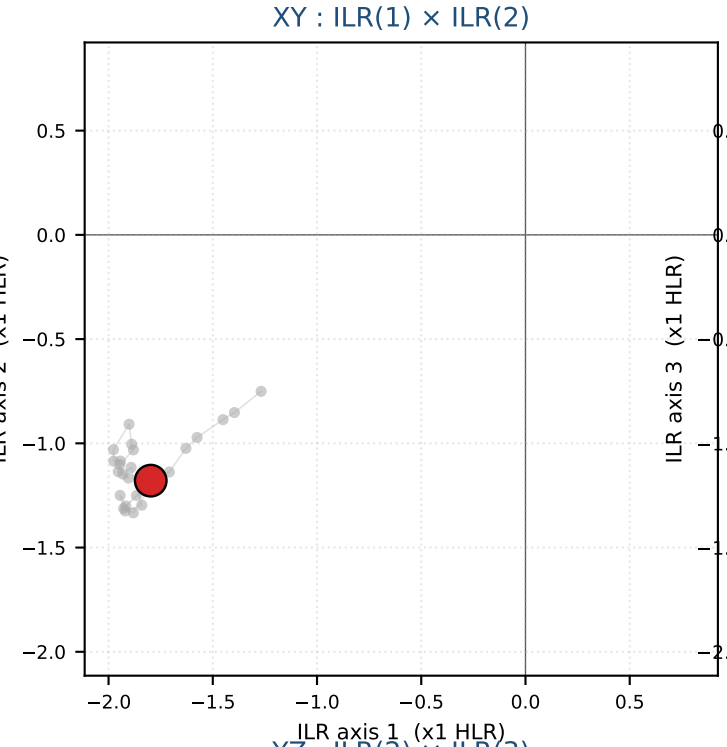
display factor: x1 (raw HLR x 1)

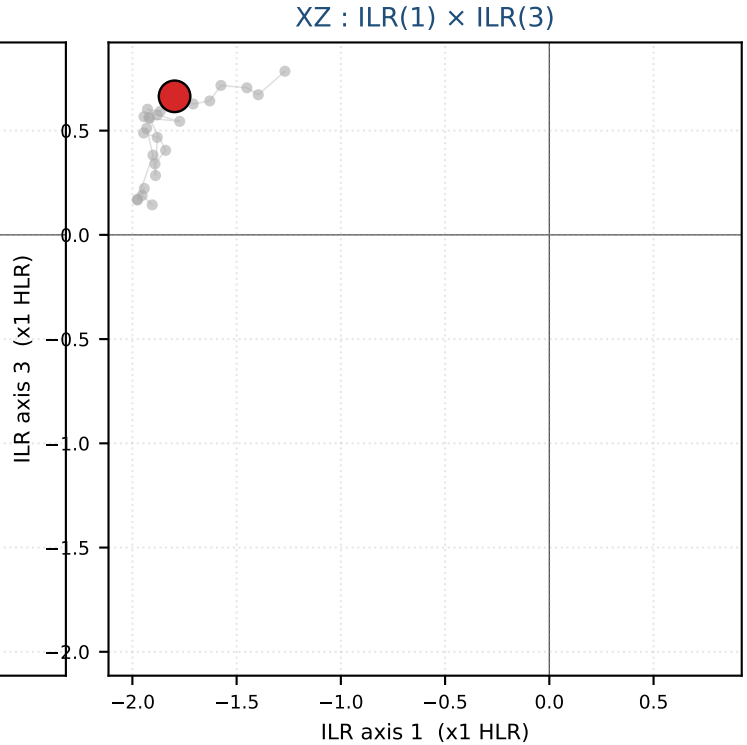
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=19 | 2019 | D=8 T=26 | level 1 first-principles

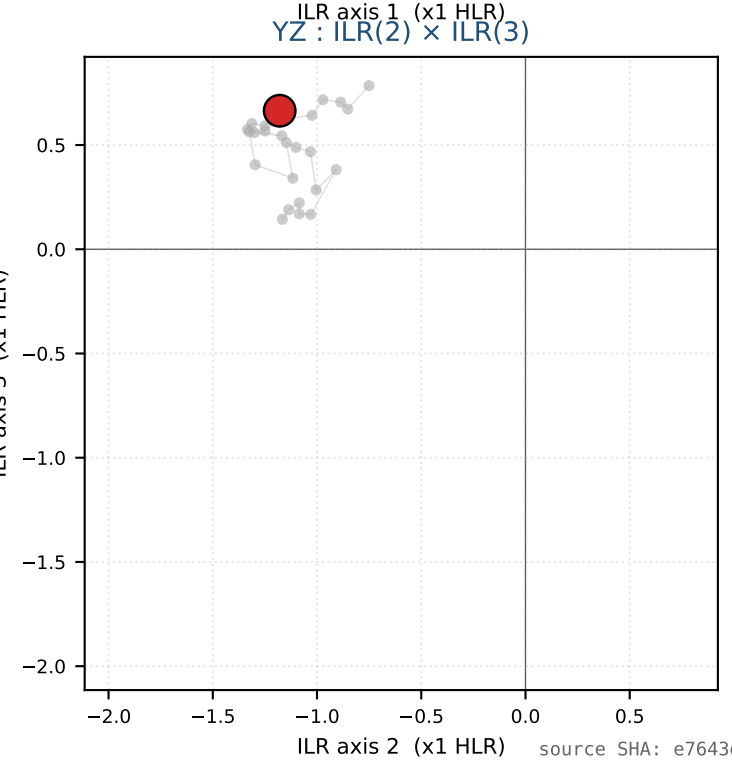
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

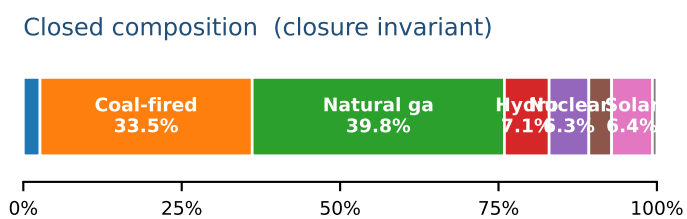




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0264	-0.876
Coal-fired	0.3349	+1.667
Natural gas	0.3981	+1.840
Hydro	0.0706	+0.110
Nuclear	0.0627	-0.009
Other fossil f	0.0359	-0.566
Solar	0.0644	+0.018
Wind	0.0071	-2.184





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

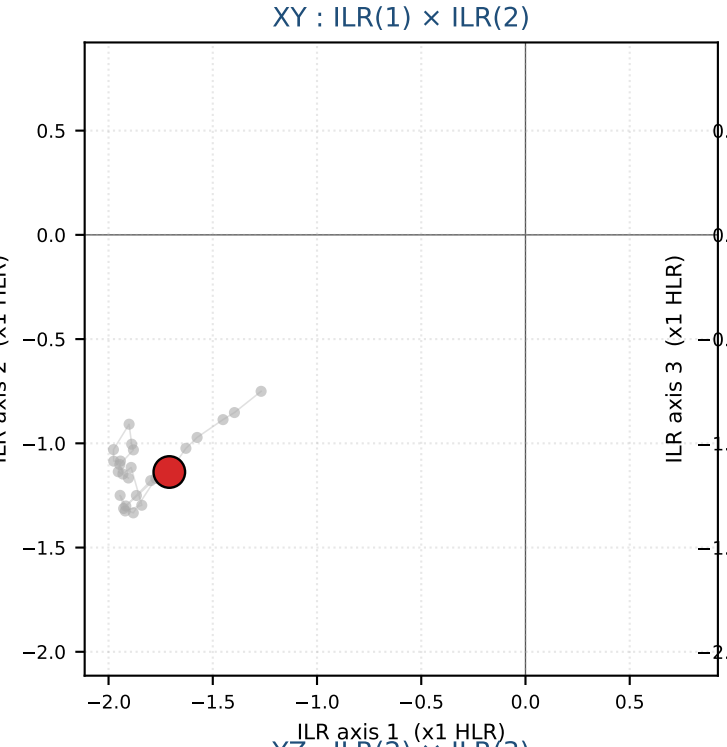
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

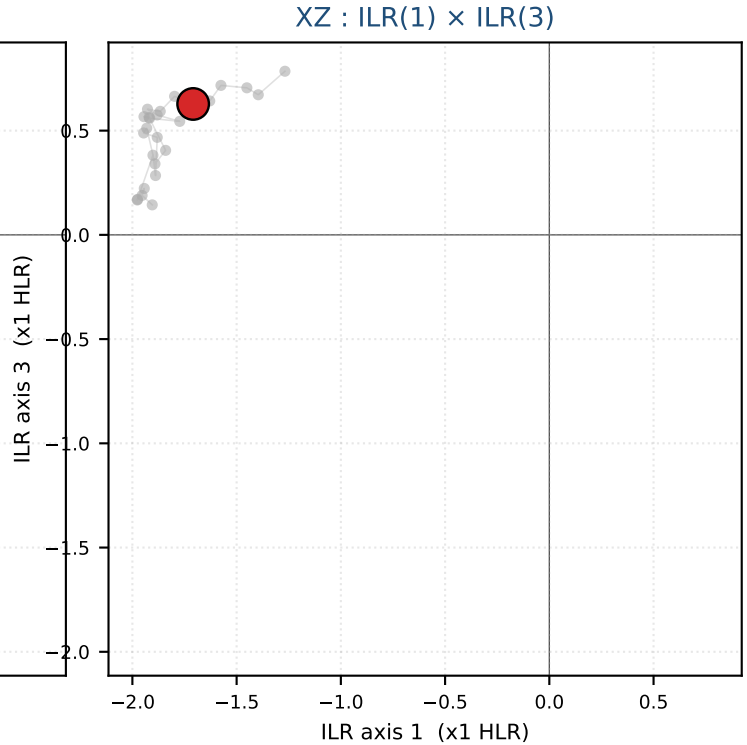
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=20 | 2020 | D=8 T=26 | level 1 first-principles

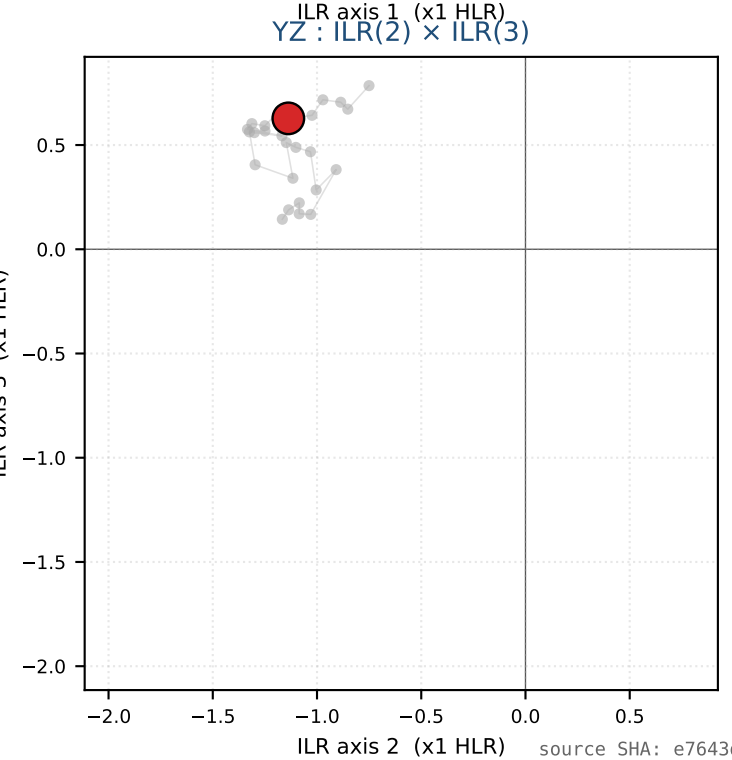
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



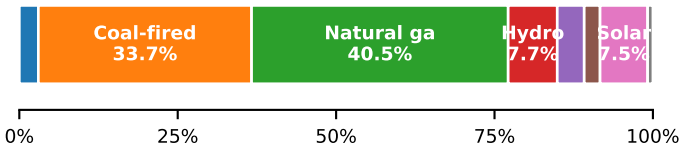


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0300	-0.720
Coal-fired	0.3365	+1.696
Natural gas	0.4050	+1.881
Hydro	0.0775	+0.227
Nuclear	0.0425	-0.373
Other fossil f	0.0250	-0.903
Solar	0.0751	+0.196
Wind	0.0083	-2.004



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

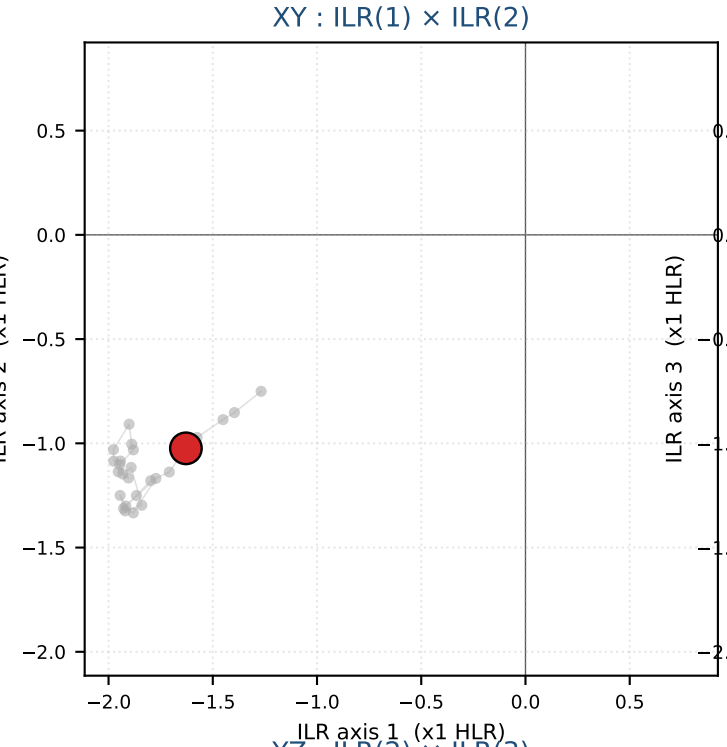
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

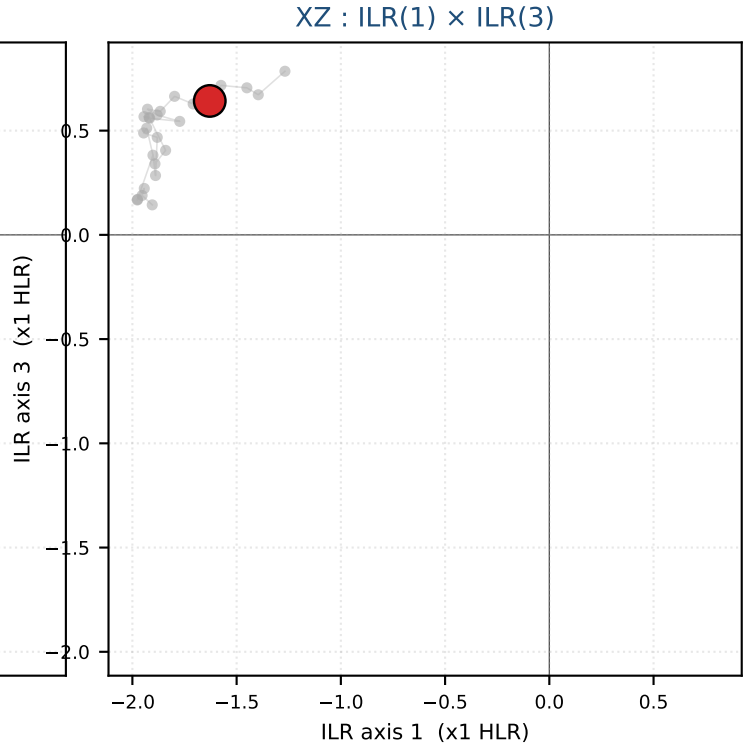
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=21 | 2021 | D=8 T=26 | level 1 first-principles

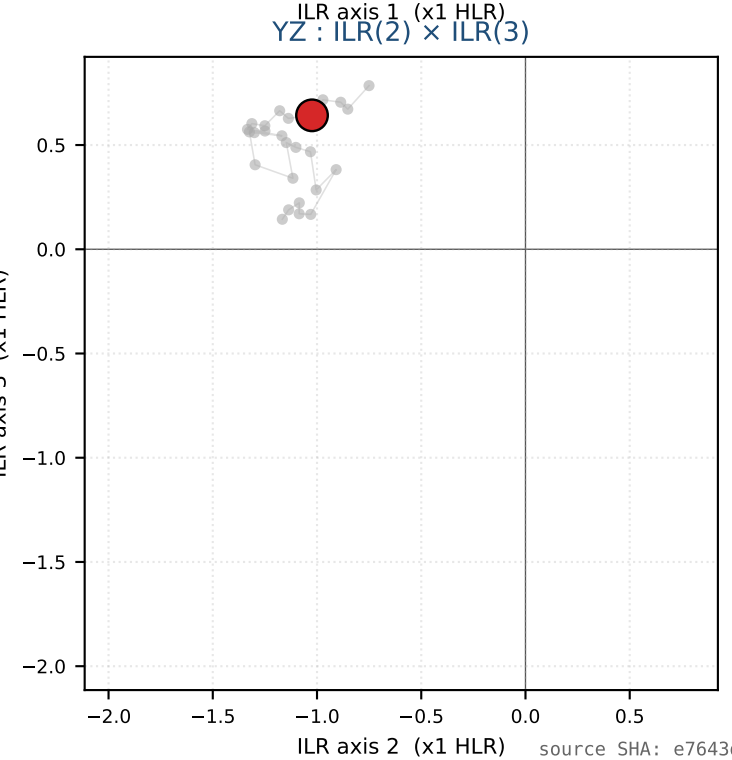
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

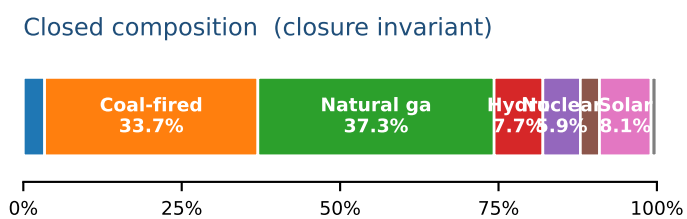




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0336	-0.698
Coal-fired	0.3366	+1.605
Natural gas	0.3729	+1.708
Hydro	0.0770	+0.130
Nuclear	0.0592	-0.133
Other fossil f	0.0302	-0.805
Solar	0.0814	+0.185
Wind	0.0092	-1.992





closure: Σ = 1.000000 OK

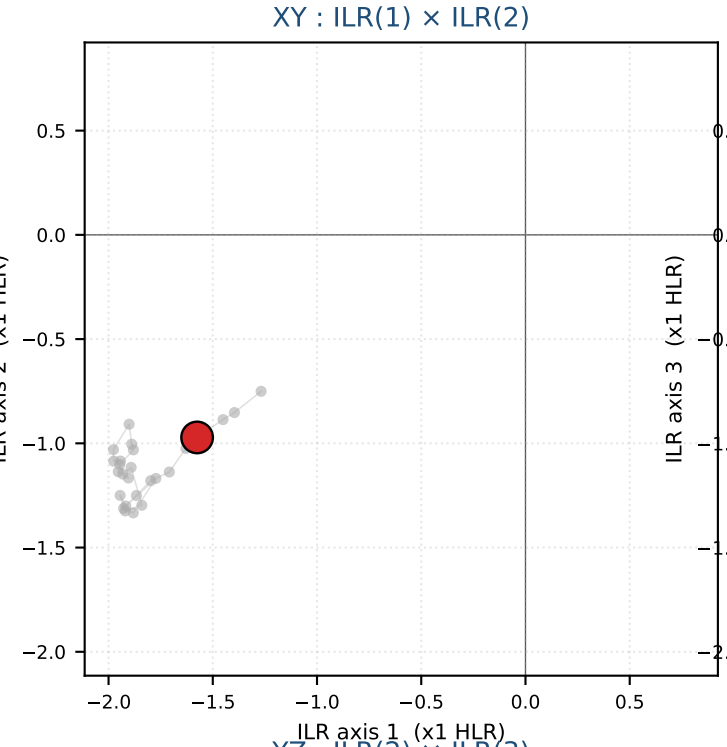
display factor: x1 (raw HLR x 1)

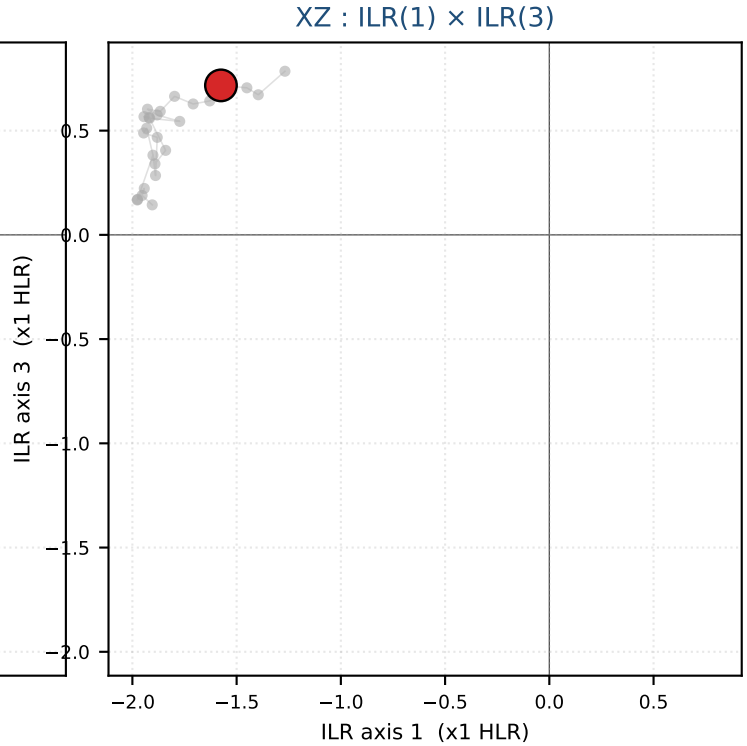
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=22 | 2022 | D=8 T=26 | level 1 first-principles

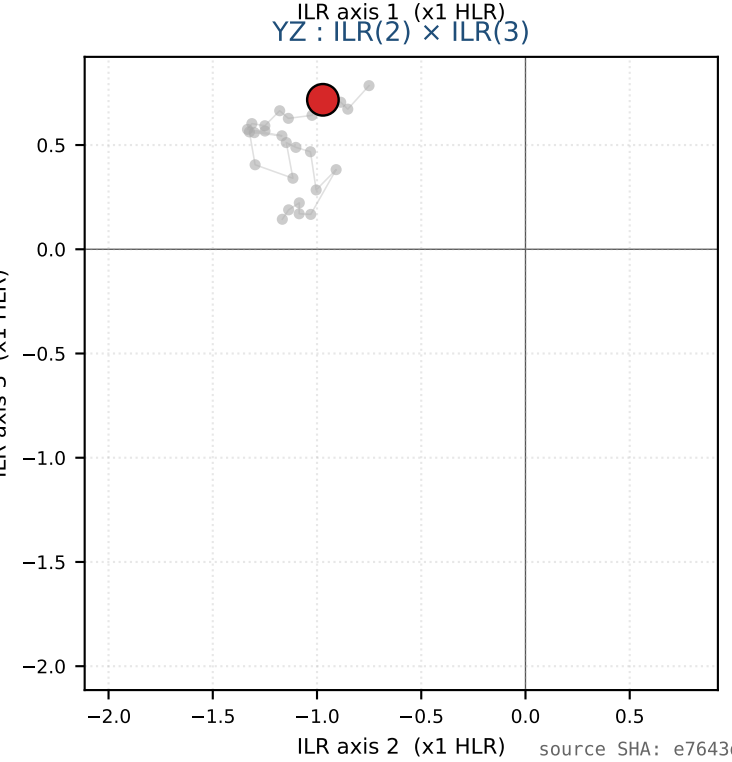
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

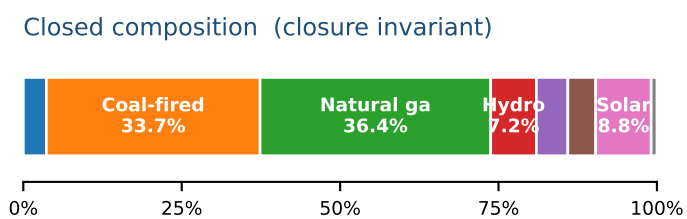




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0364	-0.651
Coal-fired	0.3373	+1.577
Natural gas	0.3641	+1.653
Hydro	0.0720	+0.032
Nuclear	0.0498	-0.337
Other fossil f	0.0440	-0.460
Solar	0.0875	+0.228
Wind	0.0091	-2.041





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

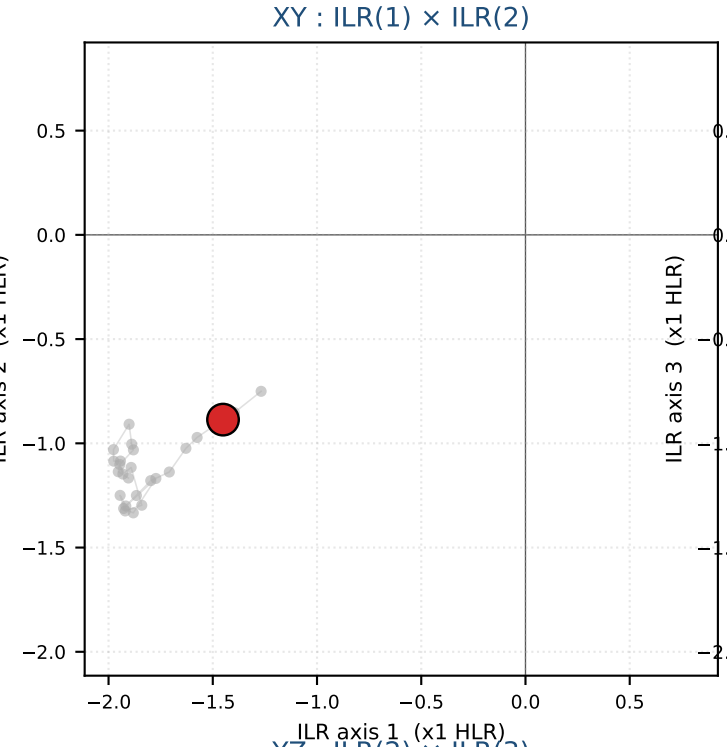
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

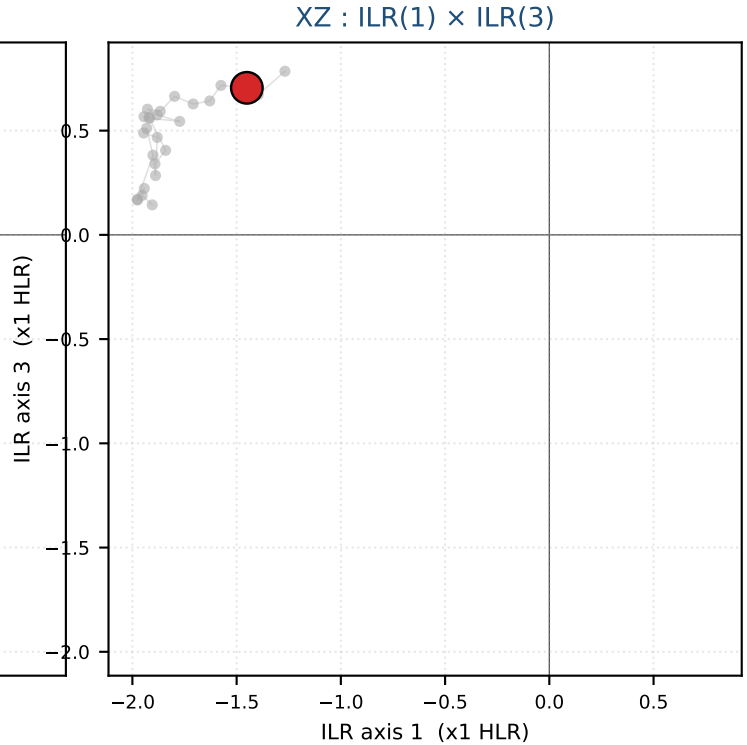
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=23 | 2023 | D=8 T=26 | level 1 first-principles

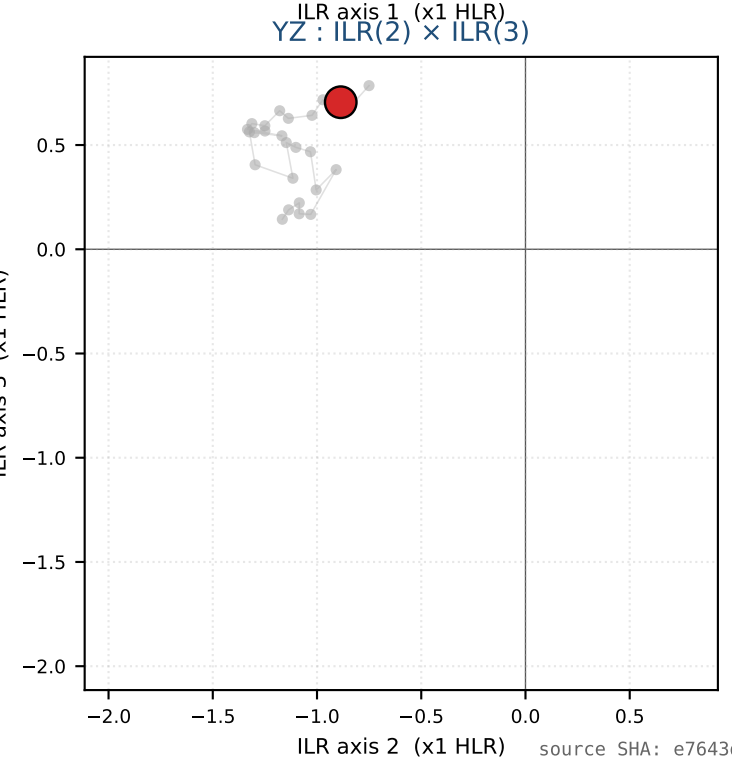
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

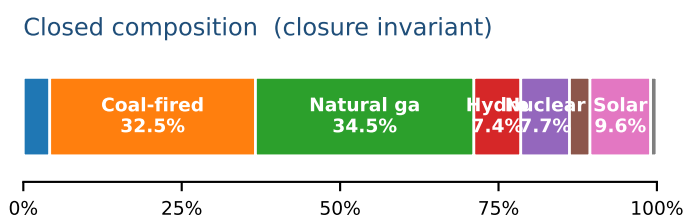




Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0417	-0.560
Coal-fired	0.3247	+1.492
Natural gas	0.3447	+1.551
Hydro	0.0740	+0.013
Nuclear	0.0769	+0.052
Other fossil f	0.0323	-0.817
Solar	0.0957	+0.270
Wind	0.0099	-2.000





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

Helmert basis loadings (top 3 carriers per axis)

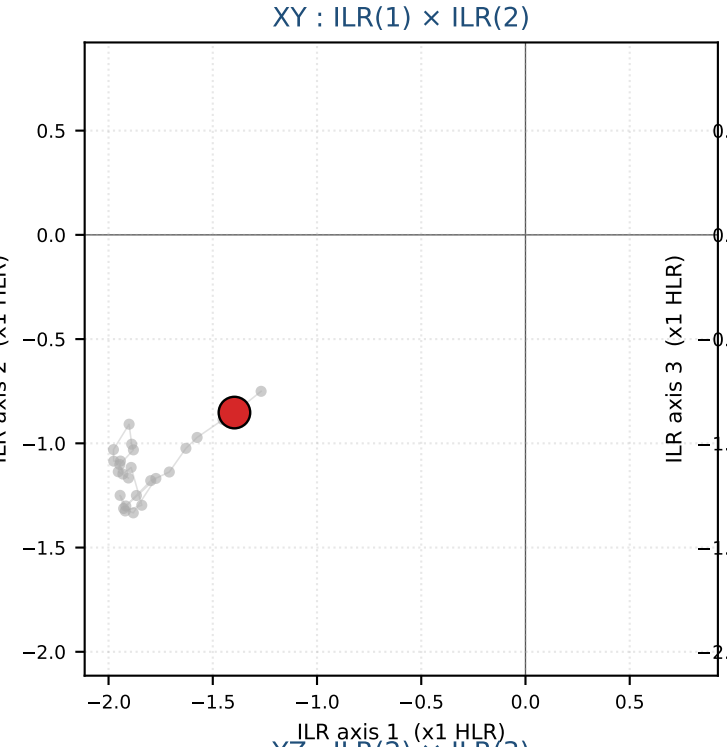
ILR(1): Biomass /+0.707 Coal-fire-0.707 Natural g+0.000

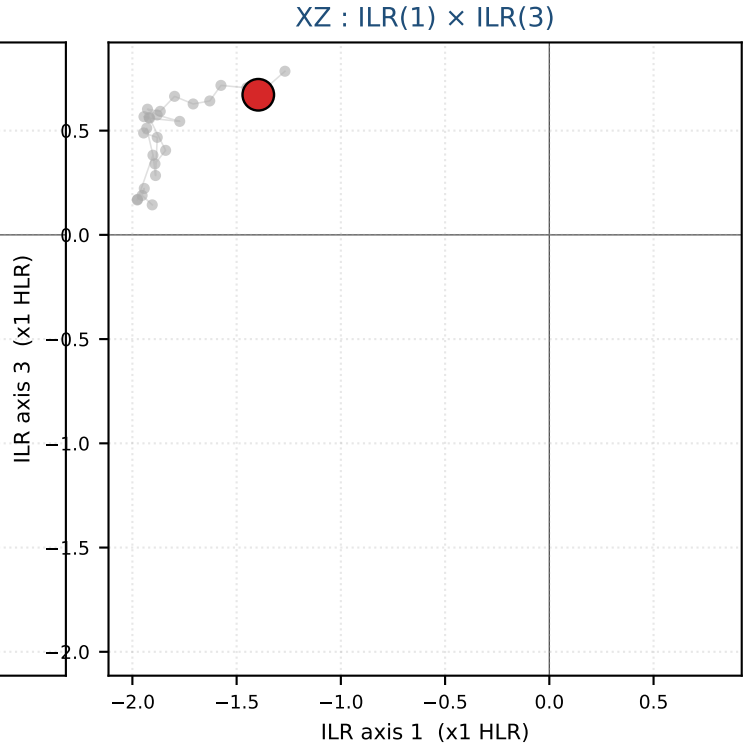
ILR(2): Natural g-0.816 Biomass /+0.408 Coal-fire+0.408

ILR(3): Hydro-0.866 Biomass /+0.289 Coal-fire+0.289

STAGE 1 v4 | t=24 | 2024 | D=8 T=26 | level 1 first-principles

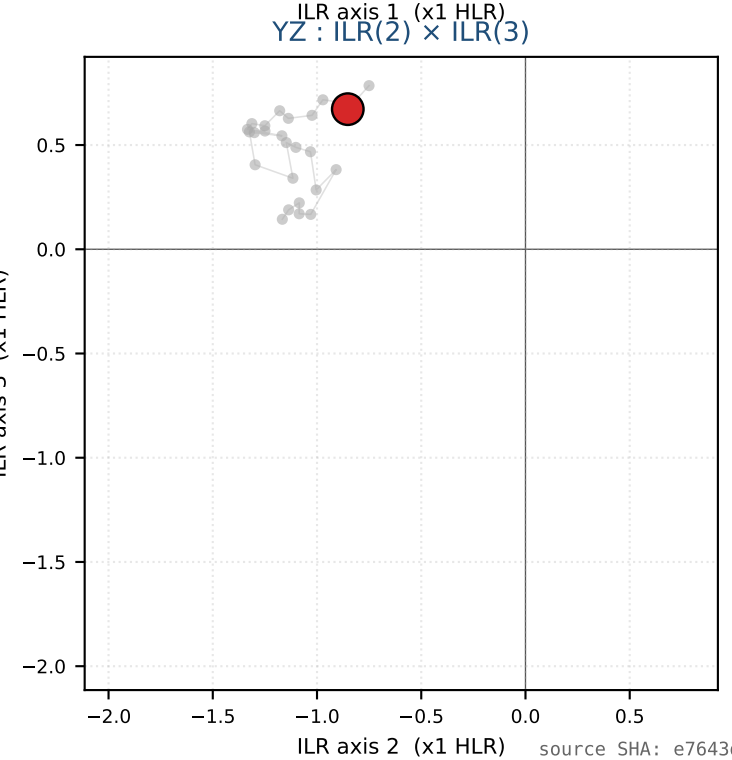
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



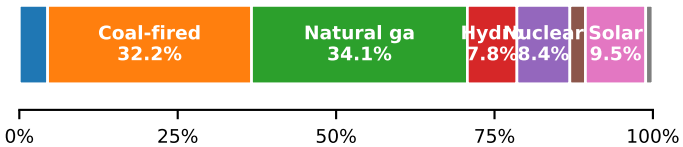


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0447	-0.497
Coal-fired	0.3219	+1.477
Natural gas	0.3407	+1.534
Hydro	0.0782	+0.062
Nuclear	0.0835	+0.128
Other fossil f	0.0246	-1.094
Solar	0.0951	+0.258
Wind	0.0114	-1.867



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

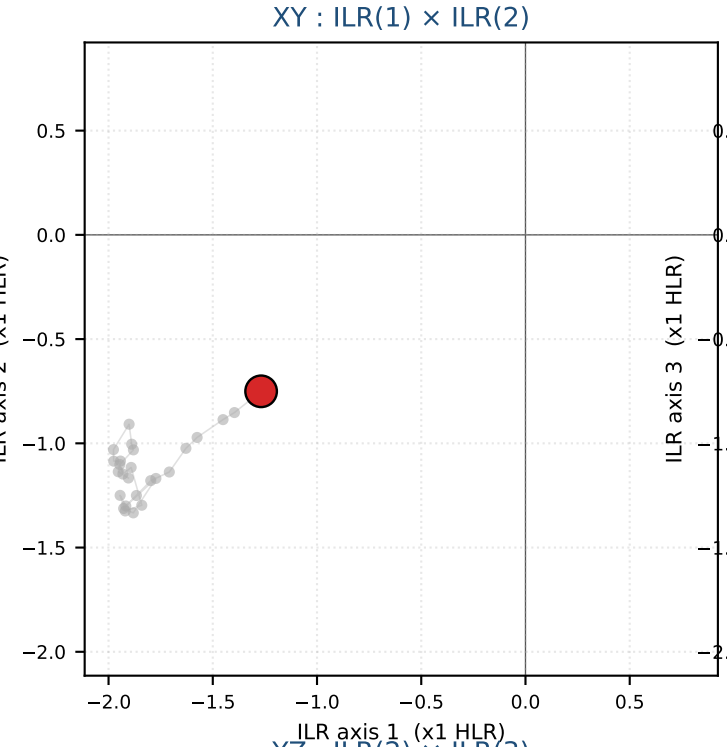
display factor: x1 (raw HLR x 1)

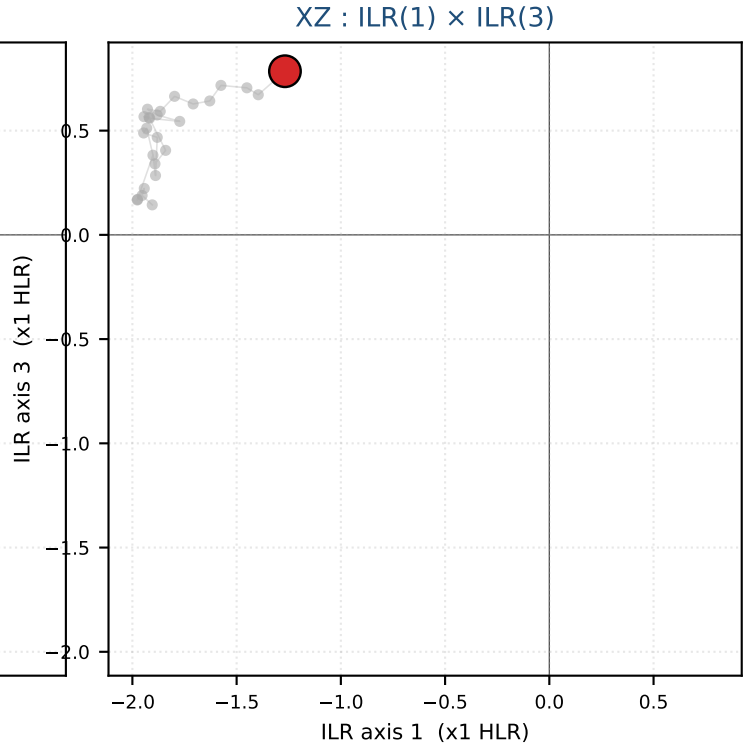
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

STAGE 1 v4 | t=25 | 2025 | D=8 T=26 | level 1 first-principles

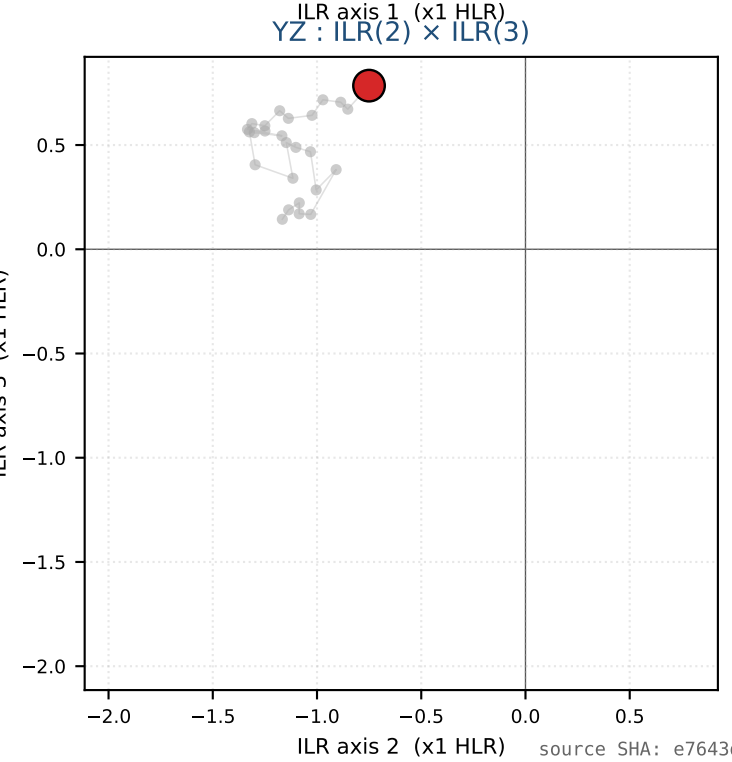
ember_jpn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



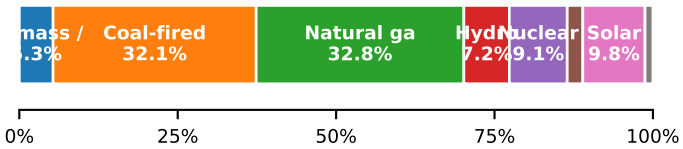


Closed composition (first principles)

carrier	closed	CLR
Biomass / biof	0.0533	-0.354
Coal-fired	0.3205	+1.440
Natural gas	0.3278	+1.462
Hydro	0.0718	-0.057
Nuclear	0.0914	+0.185
Other fossil f	0.0247	-1.125
Solar	0.0981	+0.256
Wind	0.0125	-1.806



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

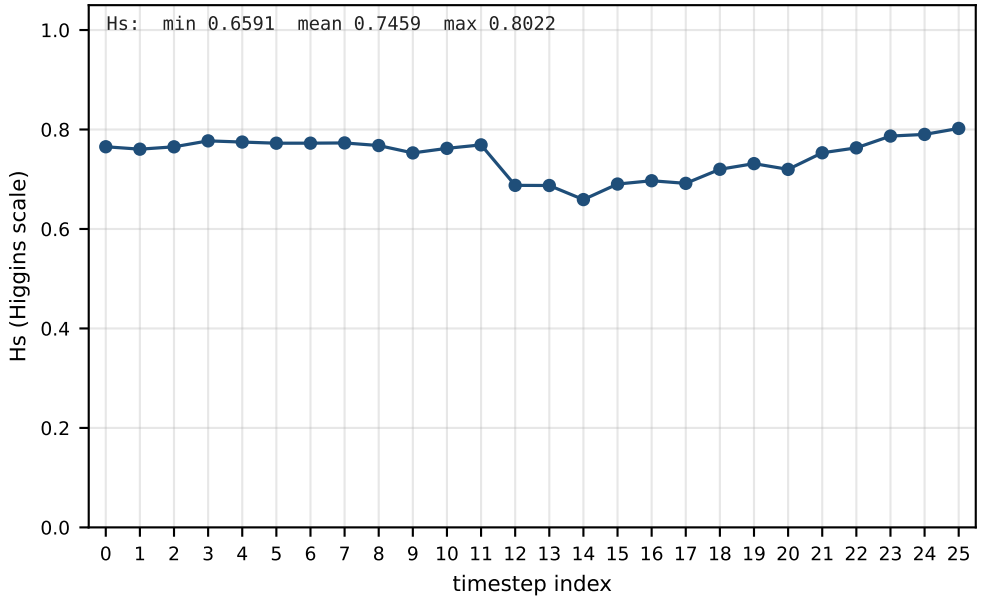
Helmert basis loadings (top 3 carriers per axis)

```
ILR(1): Biomass /+0.707  Coal-fire-0.707  Natural g+0.000
ILR(2): Natural g-0.816  Biomass /+0.408  Coal-fire+0.408
ILR(3):     Hydro-0.866  Biomass /+0.289  Coal-fire+0.289
```

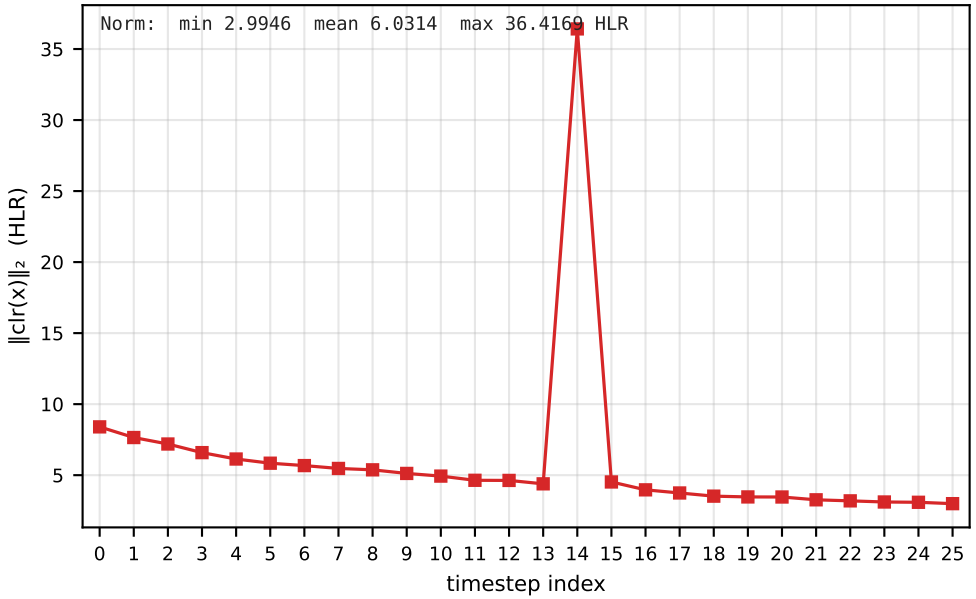
STAGE 1 SUMMARY | ember_jpn | D=8 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.3 | Schema 2.0.0 | Generated 2026-05-05T20:04:42

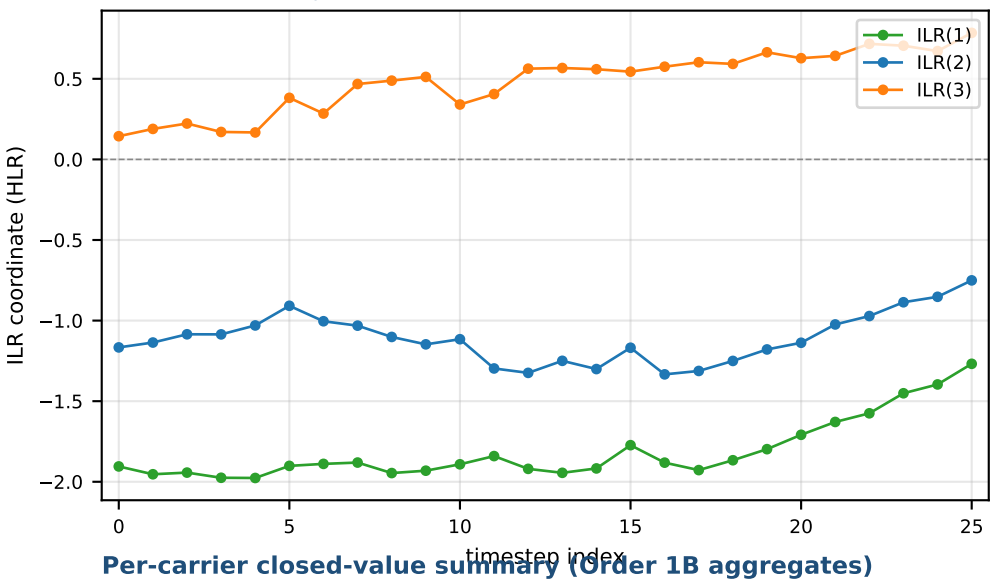
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 3.33e-16
closure status        :  OK ✓

Hs range              :  [0.6591, 0.8022]
Aitchison norm range  :  [2.9946, 36.4169] HLR
ILR(1) range          :  [-1.9765, -1.2682]
ILR(2) range          :  [-1.3336, -0.7505]
ILR(3) range          :  [+0.1439, +0.7849]

Source file           :  ember_JPN_Japan_generation_TWh.csv
Source SHA-256        :  e7643dca648ecadc9668287719d669d4...
Closed-data SHA-256   :  02f7db429142fc17907874ac6860e92b...
Content SHA-256       :  fded914144fd6a8542717c357440dce7...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Biomass / biof	0.0146	0.0243	0.0533	0.0098
1	Coal-fired	0.2166	0.2977	0.3394	0.0374
2	Natural gas	0.2195	0.3278	0.4310	0.0735
3	Hydro	0.0631	0.0740	0.0846	0.0054
4	Nuclear	0.0000	0.1360	0.2959	0.1071
5	Other fossil f	0.0246	0.1030	0.1877	0.0549
6	Solar	0.0003	0.0323	0.0981	0.0361
7	Wind	0.0001	0.0048	0.0125	0.0035